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MASTER OF SCIENCE IN MATHEMATICS

A Study of Meta-Learners and Cross-Fitting: Performance Evaluation in Diverse Experimental Settings

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Admission for Circulating the Work

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Foreword

This dissertation represents the culmination of my research into evaluating meta-learners and cross-fitting techniques for estimating the Conditional Average Treatment Effect (CATE). My primary contribution to this work has been the comparative analysis of different algorithms, aimed at determining their effectiveness across various experimental settings.

I would like to express my deepest gratitude to my supervisor, Dr. Stijn Vansteelandt, for his invaluable guidance throughout this research. His expertise and support were crucial in navigating the complexities of the topic. I also extend my thanks to my family and friends for their continuous encouragement.

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Summary

The primary goal of this thesis was to evaluate the performance of various meta-learners—specifically, the S-learner, T-learner, Inverse Probability Weighting (IPW) learner, Doubly Robust (DR) learner, X-learner, and R-learner—in estimating the Conditional Average Treatment Effect (CATE). Additionally, the research aimed to determine whether applying cross-fitting to these meta-learners could improve the accuracy and robustness of the CATE estimates across different scenarios.

The study began by establishing the foundational concepts of causal inference, with a particular focus on the potential outcomes framework. This framework forms the theoretical basis for defining causal effects, especially the CATE. To accurately estimate the CATE, several key assumptions must be met, including the Stable Unit Treatment Value Assumption (SUTVA), positivity, and conditional exchangeability. These assumptions are crucial as they enable the identification of the CATE from the observed data, ensuring that the causal effects are properly inferred.

Building on this foundation, the methodology chapter described the meta-learners employed in the study, each with its strengths and weaknesses depending on the application context. The cross-fitting algorithm was also explained in detail, highlighting its role in partitioning data into folds to reduce overfitting and enhance the reliability of estimates.

Following this, the thesis outlined the simulation design used to generate data and evaluate the meta-learners' performance. These simulations were meticulously crafted to reflect real-world scenarios, varying in sample sizes, covariate dimensions, correlation structures, noise levels, and treatment assignments. This diversity allowed for a thorough assessment of each learner's performance under different conditions. The evaluation was based on three key metrics: average mean squared error , average absolute bias, and average variance.

The results of the simulations revealed several key insights. Surprisingly, cross-fitting did not improve the accuracy of the CATE estimates; in fact, it consistently led to worse performance compared to full-sample estimates. The study also found that the performance of the meta-learners varied significantly depending on the scenario. The S-, T-, and X-learners performed best when the number of covariates was small, while the DR- and R-learners excelled in high-dimensional settings. The IPW-learner consistently performed poorly across all scenarios, leading to a recommendation against its use based on our simulation results. The S-learner, while effective in scenarios with no treatment effect, exhibited high bias and low variance across different scenarios, indicating its limited utility in detecting treatment effects, as its CATE estimates tend to be biased toward zero.

The results also indicated that as covariate correlation increased, the performance of all learners

deteriorated. In high-noise scenarios, the X- and T-learners outperformed others, while in low-noise scenarios, the R-, DR-, and X-learners were most effective. The X-learner was particularly strong in scenarios with imbalanced treatment assignments, while the R-learner excelled in cases with complex treatment effects, performing best in terms of bias.

In conclusion, this thesis provided valuable insights into the performance of meta-learners with and without cross-fitting. While cross-fitting did not improve the estimates as anticipated, the study highlighted the importance of understanding the specific characteristics of the data when choosing the appropriate method for CATE estimation.

Chapter 1

Introduction

1.1 Background

Causal Inference

Causal inference is a fundamental aspect of statistics and econometrics, focusing on identifying and understanding the causal relationships between treatments or interventions and their outcomes. For instance, in healthcare, it is crucial for determining whether a new drug is more effective than a placebo in alleviating symptoms. This analysis is vital across various fields such as medicine, economics, and social sciences, where establishing causality is key to informed decision-making. Central to causal inference is the estimation of treatment effects, often quantified through measures like the Average Treatment Effect (ATE) or the Conditional Average Treatment Effect (CATE) when accounting for covariates.

A significant challenge in causal inference is the presence of confounding variables—factors that can distort the relationship between the treatment and the outcome, leading to biased estimates. For example, consider a study that aims to evaluate the effectiveness of a new medication in reducing blood pressure. In an ideal randomized controlled trial (RCT), participants would be randomly assigned to either receive the medication or a placebo, ensuring that any differences in outcomes can be attributed to the medication itself. Randomization helps distribute confounding variables, such as age, baseline health status, and diet, evenly across treatment groups.

However, in many real-world situations, conducting an RCT may not be feasible due to ethical, financial, or logistical constraints. For instance, it might be unethical to withhold potentially beneficial treatment from a control group. As a result, researchers often rely on observational studies, where the assignment to treatment and control groups is not random. In our example,

if we were to conduct an observational study, patients who choose to take the new medication might differ from those who do not in ways that also affect blood pressure, such as their overall health consciousness or access to healthcare. These differences introduce confounding, making it challenging to isolate the effect of the medication.

To address these challenges, sophisticated statistical techniques are required to account for confounders and biases. One widely used framework is Rubin's Potential Outcomes Framework, also known as the Neyman-Rubin Causal Model [1]. This framework conceptualizes causal effects as the difference between the potential outcomes that would have been observed under each treatment condition. However, since only one of these potential outcomes is observed for each individual, researchers must estimate the unobserved outcomes in a way that accurately captures the causal effect.

Various methods have been developed to tackle confounding in causal inference. Liuyi Yao et al., in their comprehensive survey of causal inference methods, outline a wide range of approaches including re-weighting methods, stratification, matching-based methods, tree-based approaches, representation-based techniques, multi-task learning, and meta-learning methods [2]. Each method offers distinct advantages depending on the structure of the data and the research question at hand.

In this thesis, we focus on meta-learning methods, which have gained prominence in recent years for their ability to flexibly and powerfully estimate the CATE.

Meta-Learners

Building on the principles of causal inference, meta-learners have emerged as a powerful class of machine learning algorithms specifically designed to estimate the CATE with high precision. Traditional statistical approaches often face challenges when dealing with complex datasets, particularly in accurately capturing the nuanced relationships between variables that are essential for estimating causal effects. Meta-learners address these challenges by leveraging the flexibility and adaptability of modern machine learning models.

Meta-learners are a class of algorithms that combine supervised learning methods (i.e., base learners) in a specific manner while allowing the base learners to take any form. Among the most well-known meta-learners are the T-learner, S-learner, and X-learner. The T-learner operates by training separate models for the treated and control groups, with the CATE estimated as the difference between the predicted outcomes from these models. In contrast, the S-learner integrates the treatment assignment directly into a single model, using covariates and treatment indicators to estimate the CATE. The X-learner, introduced by Künzel et al. [3], enhances these approaches by initially estimating treatment effects separately and then refining these estimates through a second stage of modeling aiming to improve the accuracy in settings with heteroge-

neous treatment effects.

In addition to these meta-learners, other important methods have been developed to further improve the robustness and reliability of CATE estimates. For example, Inverse Probability Weighting (IPW) mitigates confounding by assigning weights to observations based on the inverse probability of receiving the treatment, thereby creating a pseudo-population where the treatment assignment is independent of the covariates. Doubly Robust (DR) methods take this a step further by combining IPW with outcome regression, allowing for estimates that remain robust even when the treatment assignment model or the outcome model is misspecified. Kennedy [4] explores optimal strategies for implementing these doubly robust methods, providing guidance on their practical application. Additionally, Nie and Wager [5] introduce the R-learner, which focuses on modeling residuals to flexibly estimate heterogeneous treatment effects, offering an alternative approach that could adapt to different data structures.

Through the effective use of these meta-learners, researchers can generate more precise and individualized treatment effect estimates. This capability is particularly valuable in fields like personalized medicine, economics, and public policy, where understanding the heterogeneous impacts of interventions can lead to more informed and targeted decision-making.

Cross-Fitting

Building on the use of meta-learners for estimating the CATE, it is crucial to address potential challenges such as overfitting and the need for robust estimates. To this end, sample splitting and cross-fitting are widely employed techniques in causal inference, particularly when leveraging machine learning methods.

Sample splitting is a technique in which the dataset is divided into distinct subsets or folds. One subset is used to estimate nuisance functions—functions that, while not of direct interest, are necessary for estimating the CATE. The other subset is reserved for estimating the treatment effects. This separation is designed to reduce the risk of overfitting by ensuring that the data used to estimate nuisance functions does not directly influence the estimation of treatment effects.

To further reduce variance, cross-fitting switches the roles of the data parts so that each part is used for both nuisance function estimation and causal effect estimation in different iterations. The final causal effect estimator is then computed as the median or mean of the separate effect estimates produced in each iteration.

The importance of cross-fitting in improving the performance of meta-learners is highlighted in several studies. Jacob [6] investigates the finite sample performance of cross-fitting in conjunction with meta-learners such as the doubly-robust learner, R-learner, T-learner, and X-learner, demonstrating that the effectiveness of these learners heavily depends on how the data is split

and averaged. Similarly, Okasa [7] emphasizes the role of sample splitting and cross-fitting in reducing overfitting bias, particularly in large samples, though they note that full-sample estimation may be more suitable for smaller datasets. Further, Knaus et al. [8] explore the impact of cross-fitting on causal machine learning estimators across various data generation processes (DGPs), showing that estimators employing cross-fitting consistently perform well across different scenarios, thus underscoring the robustness by this technique.

Overall, sample splitting and cross-fitting are critical tools in the causal inference toolkit, particularly when applying meta-learners to complex datasets. By systematically reducing overfitting and enhancing the robustness of estimates, these techniques play a vital role in ensuring that CATE estimates are both accurate and reliable.

1.2 Problem Statement

Building on the discussion of meta-learners and the role of cross-fitting in enhancing their performance, it becomes clear that accurate estimation of the CATE is essential for informed decision-making in various fields. While meta-learners have demonstrated significant potential in handling the complexities of causal inference, their effectiveness can be highly sensitive to several factors, such as sample size, the complexity of covariates, the presence of noise in outcomes, and the specific configurations of the underlying machine learning models.

Previous research has laid important groundwork in this area. Künzel et al. [3] have rigorously evaluated the performance of different meta-learners under varying conditions, shedding light on their strengths and potential limitations. More recently, studies by Daniel Jacob and Gabriel Okasa [6, 7] have further investigated the use of cross-fitting as a means to mitigate overfitting and improve the robustness of CATE estimates within the framework of meta-learners. These studies underscore the potential benefits of cross-fitting but also highlight that its impact may vary depending on the context in which it is applied.

Despite these advances, there remains a considerable gap in understanding the specific conditions under which cross-fitting most effectively enhances the accuracy and reliability of CATE estimates. For instance, the interaction between cross-fitting and factors such as varying sample sizes, covariate space complexity, levels of outcome noise, and the configurations of base learners remains underexplored. Moreover, other critical factors, including the density of correlation matrices, the degree of confounding and the balance of propensity scores, warrant further investigation to fully grasp how they influence the performance of cross-fitting in meta-learner applications.

This thesis seeks to fill these gaps by systematically evaluating the influence of cross-fitting on the performance of meta-learners across a wide range of scenarios. By examining these factors

in depth, this research aims to provide a comprehensive understanding of when and how cross-fitting can be leveraged to enhance the accuracy and robustness of CATE estimates, ultimately offering practical guidance for its application in diverse real-world settings.

1.3 Objectives of the Research

Building on the identified research gaps, the primary objective of this study is to systematically evaluate the performance of six widely recognized meta-learners—namely, the S-learner, T-learner, Inverse Probability Weighting (IPW) learner, R-learner, Doubly Robust (DR) learner, and X-learner—in estimating the CATE. A central focus of this evaluation will be the application of cross-fitting to these meta-learners, aiming to determine how effectively cross-fitting improves the accuracy and robustness of CATE estimates across a variety of scenarios.

Additionally, this research seeks to compare the performance of these different learners against each other, providing a comprehensive understanding of their relative strengths and weaknesses under various conditions.

To achieve these objectives, we will conduct a series of experiments that explore the influence of several key factors on the performance of meta-learners. One critical factor under investigation is sample size. We will assess how different dataset sizes impact the effectiveness of cross-fitting, particularly in enhancing the reliability of CATE estimates in both small and large samples.

Another significant aspect of this research is the size of the covariate space. We will examine how the number of covariates in the data affects the performance of meta-learners, with and without cross-fitting. This consideration is particularly crucial in real-world applications where high-dimensional data poses challenges for causal inference.

Additionally, this study will evaluate the role of outcome noise in the accuracy of CATE estimation. We will analyze how varying levels of noise in the outcome variable influence the estimates produced by meta-learners and whether cross-fitting effectively mitigates the negative impact of such noise.

The base learners used in this research will be Random Forest models. We will explore how varying the settings of these Random Forest models—such as the number of trees, the depth of trees, and other hyperparameters—affects the performance of CATE estimation. This will help us understand the interaction between base learner configurations and the application of cross-fitting. Further, we will investigate the impact of the density of correlation matrices within the covariates. By varying this density, we aim to understand how the relationships among covariates influence the performance of cross-fitting in improving CATE estimates.

The level of confounding in the data will be a key focus of this study. We will evaluate how varying degrees of confounding influence the effectiveness of cross-fitting, particularly in reducing bias and improving the precision of estimates. Additionally, this research will examine the balance of propensity scores across different scenarios, assessing how balance or imbalance impacts the robustness of CATE estimates with cross-fitting. Various configurations of the cross-fitting algorithm will also be explored.

Through these comprehensive investigations, this research aims to provide deeper insights into the conditions under which cross-fitting enhances or hinders the performance of meta-learners. The ultimate goal is to offer practical guidelines for researchers and practitioners, ensuring that the application of cross-fitting in diverse real-world settings leads to accurate and reliable CATE estimates.

1.4 Structure of the Thesis

This thesis is organized into three main chapters, each building upon the previous one to provide a comprehensive understanding of the research undertaken.

The second chapter, **Framework**, explores the foundational concepts underlying the research. It begins with an explanation of causal inference, introducing key notations and the potential outcomes model. The chapter then outlines the fundamental assumptions necessary for causal inference, ensuring the reader grasps the basics required for the more advanced discussions later in the thesis.

The third chapter, **Methodology**, details the methods used in this research. It starts with an explanation of cross-fitting algorithms, followed by a discussion of the various meta-learners evaluated in the study, including the S-learner, T-learner, IPW-learner, R-learner, DR-learner, and X-learner. The chapter also describes the simulation procedures, explaining how the data is generated and the different scenarios considered in the study. This thorough explanation of the methodology provides the context needed for interpreting the results presented in the following chapter.

The final chapter, **Results**, presents the research findings, analyzing the performance of the meta-learners across different scenarios with and without the application of cross-fitting. The results are discussed in depth, highlighting the conditions under which cross-fitting enhances or hinders the accuracy and robustness of CATE estimates.

This structure ensures a logical progression from the introduction of key concepts, through a detailed methodology, to the presentation and analysis of results, ultimately leading to a well-supported conclusion.

Chapter 2

Framework

This chapter establishes the theoretical foundation for the research conducted in this thesis, introducing the essential concepts and frameworks necessary for understanding the methodologies and analyses that follow. It begins with an in-depth exploration of the causal inference framework, followed by a discussion of the critical assumptions required for reliable causal inference.

Causal inference is central to this research, offering the tools and methodologies necessary to identify and estimate the effects of treatments or interventions. This section delves into the foundational concepts of causal inference, with a particular emphasis on the Neyman-Rubin potential outcomes framework [1], a widely recognized approach for systematically examining causal relationships and treatment effects. The discussion is further informed by Neal's "Introduction to Causal Inference" [9] and Hernan and Robins' "Causal Inference: What If" [10].

2.1 Causation vs. Correlation

A critical distinction in causal inference is the difference between causation and correlation. Correlation indicates a statistical association between two variables, but it does not imply that one variable directly causes changes in the other. Causation, on the other hand, suggests a direct influence, where a change in one variable is responsible for a change in another. This distinction is fundamental in fields such as healthcare, economics, and social sciences, where determining the effectiveness of interventions is essential.

To illustrate the importance of distinguishing between correlation and causation, consider the following example: Suppose we observe a positive correlation between ice cream sales and drowning incidents. It might be tempting to infer a causal relationship between these two vari-

ables, but this would be incorrect. The increase in both ice cream sales and drowning incidents is actually due to a third variable—temperature. During the summer months, higher temperatures lead to increased ice cream consumption and more frequent swimming, which in turn raises the risk of drowning. This example underscores the importance of identifying and accounting for confounding variables in data analysis to avoid erroneous causal inferences.

2.2 Notation

Before delving into the identification of causal effects, it is essential to establish the notation that will be used throughout this dissertation. The Neyman-Rubin potential outcome framework forms the basis for our exploration of causal relationships and treatment effects, enabling us to systematically examine the potential outcomes that individuals could experience under different treatment scenarios. The following notations will be consistently employed in our analysis.

Covariates

Covariates, also known as features or independent variables, are the predictors used to explain or forecast the outcome variable. In our dataset, covariates are represented by the matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, where n denotes the number of observations and p represents the number of covariates.

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix}$$

In this representation, each row corresponds to an individual observation, while each column represents a specific covariate.

Outcome

The outcome variable, also referred to as the dependent or response variable, is the primary variable of interest that we aim to predict or explain using the covariates. The outcomes in our dataset are typically represented as a vector $\mathbf{Y} \in \mathbb{R}^n$.

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

In this representation, y_i denotes the outcome for the i-th observation.

Treatment Variable

The treatment variable, also known as the exposure or intervention variable, represents the condition or treatment applied to individuals in an observational or experimental study. In causal inference, the treatment variable is essential for analyzing treatment effects and estimating causal relationships between interventions and outcomes. Treatment assignments are typically represented as a vector $\mathbf{T} \in \mathbb{R}^n$:

$$\mathbf{T} = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_n \end{bmatrix}$$

In this representation, t_i denotes the treatment assignment for the i-th observation. For binary treatments, t_i usually takes values of 0 or 1, indicating the control and treatment groups, respectively.

Observed Data

The observed data $D = (\mathbf{X}, \mathbf{Y}, \mathbf{T})$ consists of the covariates, outcomes, and treatment assignments for each observation in the dataset.

$$\mathbf{D} = \begin{bmatrix} x_{11} & \cdots & x_{1p} & y_1 & t_1 \\ \vdots & \ddots & \vdots & \vdots & \vdots \\ x_{n1} & \cdots & x_{np} & y_n & t_n \end{bmatrix} = \begin{bmatrix} \mathbf{d}_1 \\ \vdots \\ \mathbf{d}_n \end{bmatrix}$$

This matrix serves as the foundation for our analysis, enabling the estimation of causal effects and the modeling of relationships between covariates, treatments, and outcomes.

Potential Outcomes

Within the Neyman-Rubin potential outcome framework, potential outcomes represent the outcomes that would have been observed under different treatment conditions. These outcomes are denoted by $\mathbf{Y}^t \in \mathbb{R}^n$, where t represents the treatment condition. For binary treatments (e.g., treatment vs. control), t typically takes values 0 or 1.

$$\mathbf{Y}^t = \begin{bmatrix} y_1^t \\ y_2^t \\ \vdots \\ y_n^t \end{bmatrix}$$

In this context, y_i^t represents the potential outcome for the i-th observation under the treatment condition t.

Response Functions

Response functions describe the relationship between covariates and outcomes under different treatment conditions. Specifically, we define the following response functions:

- $\mu(\mathbf{x}) = \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x})$: Represents the expected outcome given covariates $\mathbf{x}$.
- $\mu(\mathbf{x}, t) = \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x}, T = t)$: Represents the expected outcome given covariates $\mathbf{x}$ and treatment t.
- $\mu_0(\mathbf{x}) = \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x}, T = 0)$: Represents the expected outcome under the control condition given covariates $\mathbf{x}$.
- $\mu_1(\mathbf{x}) = \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x}, T = 1)$: Represents the expected outcome under the treatment condition given covariates $\mathbf{x}$.

Propensity Score Function

The propensity score function is defined as

$$e(\mathbf{x}) = P(T = 1 \mid \mathbf{X} = \mathbf{x}),$$

representing the probability of receiving the treatment given the covariates $\mathbf{x}$.

Individual Treatment Effect (ITE)

The individual treatment effect (ITE) quantifies the causal effect of the treatment for a specific individual. It is defined as:

$$ITE_i = Y_i^1 - Y_i^0$$

where Y_i^1 denotes the potential outcome for individual i under the treatment condition, and Y_i^0 represents the potential outcome under the control condition.

Average Treatment Effect (ATE)

The average treatment effect (ATE) represents the expected difference in outcomes between the treated and control groups across the population. It is mathematically defined as:

$$ATE = E(Y^1 - Y^0)$$

where Y^1 and Y^0 are the potential outcomes under the treatment and control conditions, respectively. The ATE provides a summary measure of the causal effect of the treatment at the population level.

Conditional Average Treatment Effect (CATE)

The conditional average treatment effect (CATE), denoted by $\tau(\mathbf{x})$, quantifies the average treatment effect for individuals with specific covariates $\mathbf{x}$. It is defined as:

$$\tau(\mathbf{x}) = E(Y^1 - Y^0 \mid \mathbf{X} = \mathbf{x})$$

The CATE provides a more nuanced understanding of the treatment effect by considering the variability of effects across different subpopulations characterized by their covariates.

Having established the notation and key concepts used in our analysis, we now turn to the fundamental problem of causal inference.

2.3 Fundamental Problem of Causal Inference

A central challenge in causal inference is that, for each individual, we can observe only one of the potential outcomes. For instance, if an individual receives a treatment, we observe their outcome under that treatment but do not observe what their outcome would have been without the treatment. This limitation makes it inherently difficult to directly measure the causal effect.

Consider the following example: Suppose we have 20 subjects, each suffering from a headache. Some of these subjects take a potion, while others do not. One hour later, we observe whether each subject's headache has disappeared. In this scenario, the treatment variable T is dichotomous ($T = 1$ if the potion was taken, $T = 0$ if it was not), and the outcome variable Y is also dichotomous ($Y = 1$ if the headache disappeared, $Y = 0$ if it did not). The data for this scenario can be summarized in Table 2.1.

In this example, the treatment T has a causal effect on an individual's outcome Y if $Y^1 \neq Y^0$ for that individual. However, we can only observe one of the potential outcomes for each person. For instance, if Frank did not take the potion ($T = 0$) and his headache disappeared ($Y = 1$), we know $Y^0 = 1$ but not Y^1. This limitation exemplifies the fundamental problem of causal inference, as it is impossible to directly observe the individual treatment effect $Y^1 - Y^0$.

Given this challenge, rather than attempting to infer individual treatment effects, we often focus on population-level measures, such as the average treatment effect (ATE), defined as $E(Y^1 - Y^0)$. Under certain assumptions, the ATE can be identified from data, providing a useful summary measure of the treatment effect across the entire population. However, this aggregate measure may obscure important variations in treatment effects across different subgroups.

To capture these variations, we look at the Conditional Average Treatment Effect (CATE),

	T	Y	Y^0	Y^1
Annelise	0	0	0	?
Wes	0	1	1	?
Laurel	0	0	0	?
Connor	0	0	0	?
Michaela	1	0	?	0
Asher	1	0	?	0
Oliver	1	0	?	0
Bonnie	1	1	?	1
Frank	0	1	1	?
Sam	0	1	1	?
Gabriel	0	0	0	?
Nate	1	1	?	1
Rebecca	1	1	?	1
Tegan	1	1	?	1
Emmett	1	1	?	1
Vivian	1	1	?	1
Jorge	1	1	?	1
Pam	1	0	?	0
Isaac	1	0	?	0
Simon	1	0	?	0

Table 2.1: Observed and counterfactual data

which estimates the treatment effect based on specific covariates. The CATE provides a deeper understanding by accounting for differences in how treatment effects may vary among different groups of individuals. In the following sections, we will explore the assumptions necessary for identifying the CATE and discuss their significance in causal inference.

2.4 Assumptions for Causal Inference

Accurately estimating the Conditional Average Treatment Effect (CATE) in causal inference relies on several key assumptions. These assumptions are crucial for drawing valid causal conclusions from the data. In the following sections, we will explore each of these assumptions in detail, discussing their importance and role in enabling the identification and estimation of the CATE.

SUTVA (Stable Unit Treatment Value Assumption)

The Stable Unit Treatment Value Assumption (SUTVA) is an important assumption in causal inference, comprising two key components: no interference and consistency.

The **no interference assumption** asserts that the treatment assigned to one individual does not

influence the potential outcomes of other individuals. Formally, this is expressed as:

$$Y_i^{t_1,t_2,\ldots,t_n} = Y_i^{t_i}$$

for all i, where $Y_i^{t_i}$ represents the potential outcome for individual i given their own treatment t_i, and $Y_i^{t_1,t_2,\ldots,t_n}$ represents the potential outcome considering the treatment assignments of all individuals. This assumption ensures that each individual's outcome is determined solely by their own treatment, not by the treatments assigned to others.

To illustrate what it would mean for this assumption to be violated, consider a study on the effectiveness of a vaccination program in a school. If many students receive the vaccine, the spread of the disease may be reduced, indirectly protecting even those who did not receive the vaccine. In this case, the potential outcome for a non-vaccinated student could be influenced by the vaccination status of others, violating the no interference assumption.

The **consistency assumption** posits that the observed outcome for an individual under the treatment they actually received is the same as the potential outcome under that treatment. Formally, this is expressed as:

$$Y_i = Y_i^{T_i}$$

where Y_i is the observed outcome for individual i, and $Y_i^{T_i}$ is the potential outcome under the observed treatment T_i. This implies that if an individual receives treatment $T_i = t$, the observed outcome Y_i should be identical to the potential outcome Y_i^t.

An example of where this assumption might be violated is in a study where the treatment is defined as "move to a new city" ($T = 1$). This treatment specification does not capture the differences between various types of cities. For instance, moving to a bustling metropolis may result in $Y = 1$ (satisfaction) for someone who enjoys city life, while moving to a quiet town might result in $Y = 0$ (dissatisfaction) for the same person. Since both scenarios fall under the same treatment $T = 1$ but lead to different outcomes, Y^1 is not well-defined, violating the consistency assumption.

SUTVA combines these two assumptions, ensuring that:

1. The potential outcomes for any individual are not affected by the treatments assigned to other individuals (no interference).

2. The observed outcome for any individual under a given treatment is the same as their potential outcome under that treatment (consistency).

Formally, SUTVA can be expressed as:

$$Y_i = Y_i^{T_i} \quad \text{and} \quad Y_i^{T_1,T_2,\ldots,T_n} = Y_i^{T_i}$$

for all i.

SUTVA is essential for identifying causal effects because it ensures that differences in outcomes can be attributed solely to differences in treatments, without concerns about cross-interference or inconsistencies in potential outcomes. By assuming SUTVA, we ensure that:

$$E[Y^1 \mid T = 1, \mathbf{X} = \mathbf{x}] = E[Y \mid T = 1, \mathbf{X} = \mathbf{x}]$$

and

$$E[Y^0 \mid T = 0, \mathbf{X} = \mathbf{x}] = E[Y \mid T = 0, \mathbf{X} = \mathbf{x}].$$

The importance of this calculation will become clearer as we delve into the other necessary assumptions for identifying causal effects.

Positivity / Overlap

Another critical assumption in causal inference is positivity, also known as the overlap assumption. Positivity requires that every individual in the population has a non-zero probability of receiving each level of the treatment. Formally, for all values of the covariates $\mathbf{X}$ in the population of interest, this can be expressed as:

$$0 < P(T = 1 \mid \mathbf{X} = \mathbf{x}) < 1 \quad \text{for all } \mathbf{x}.$$

Positivity is essential because it ensures that there are both treated and control individuals for every covariate pattern, allowing for meaningful comparisons between these groups. Maintaining positivity can be particularly challenging in high-dimensional settings where many covariates are considered. As the dimensionality increases, the data can become sparse, making it more likely that some covariate patterns will not have both treated and control individuals. Ensuring positivity often requires careful study design and may involve restricting the analysis to regions of the covariate space where sufficient overlap exists.

The importance of the positivity assumption can be illustrated mathematically as:

$$E[Y \mid T = 1, \mathbf{X} = \mathbf{x}] = \int_{-\infty}^{+\infty} y \cdot \frac{P(Y = y, T = 1, \mathbf{X} = \mathbf{x})}{P(T = 1 \mid \mathbf{X} = \mathbf{x}) \cdot P(\mathbf{X} = \mathbf{x})} \, dy,$$

and similarly for $T = 0$:

$$E[Y \mid T = 0, \mathbf{X} = \mathbf{x}] = \int_{-\infty}^{+\infty} y \cdot \frac{P(Y = y, T = 0, \mathbf{X} = \mathbf{x})}{P(T = 0 \mid \mathbf{X} = \mathbf{x}) \cdot P(\mathbf{X} = \mathbf{x})} \, dy,$$

where we have applied Bayes' rule. If $P(T = 1 \mid \mathbf{X} = \mathbf{x}) = 0$ or $P(T = 0 \mid \mathbf{X} = \mathbf{x}) = 0$, it becomes impossible to estimate these quantities because there are no observations for one of the treatment groups, leading to a division by zero and making the CATE undefined for that covariate pattern.

Here is an example that illustrates what it would mean for this assumption to be violated. Consider a study evaluating the effect of a new drug on a specific health outcome, where the treatment $T = 1$ is receiving the drug, and $T = 0$ is not receiving it. If the drug is only prescribed to patients with severe symptoms, and no patients with mild symptoms ever receive the drug, then the probability $P(T = 1 \mid \mathbf{X} = \text{mild symptoms patient})$ for mild symptom patients is zero. This situation violates the positivity assumption because it becomes impossible to compare treated and untreated patients within the group of mild symptom patients, as there are no observations of treated patients in this group. Consequently, estimating the causal effect for this covariate pattern is not feasible.

Before introducing the final assumption, which will enable us to identify the CATE from the data, we need to first discuss the concept of confounding.

Confounding and Causal Diagrams

Confounding is a critical concept in causal inference, where a third variable, known as a confounder, influences both the treatment and the outcome, potentially distorting the estimation of the true causal effect. Understanding and addressing confounding is essential for accurate causal inference. One effective way to visualize and reason about confounding is through the use of causal diagrams, specifically causal directed acyclic graphs (causal DAGs).

Causal diagrams are graphical representations that illustrate the relationships between variables in a causal system. In a causal DAG, nodes represent variables, and directed edges (arrows) indicate possible causal relationships. Importantly, causal DAGs are acyclic, meaning they do not allow for any circular relationships where a variable could indirectly influence itself through a sequence of other variables.

Consider a simple causal DAG with three variables: the treatment T, the outcome Y, and a potential confounder X (see Figure 2.1).

In this diagram:

- $T \to Y$ indicates that the treatment T could have a causal effect on the outcome Y.

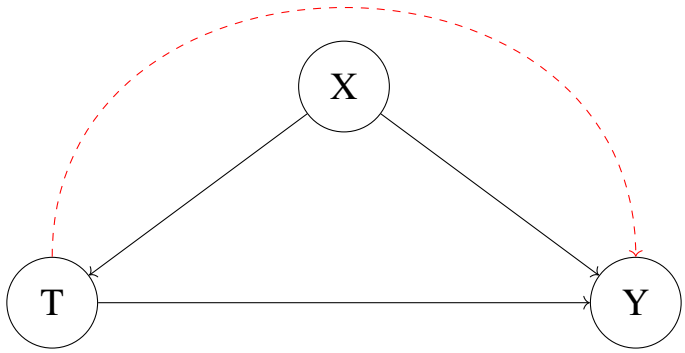

Figure 2.1: A simple causal DAG illustrating a confounder X affecting both treatment T and outcome Y.

- $X \to T$ and $X \to Y$ indicate that the confounder X could influence both the treatment T and the outcome Y.

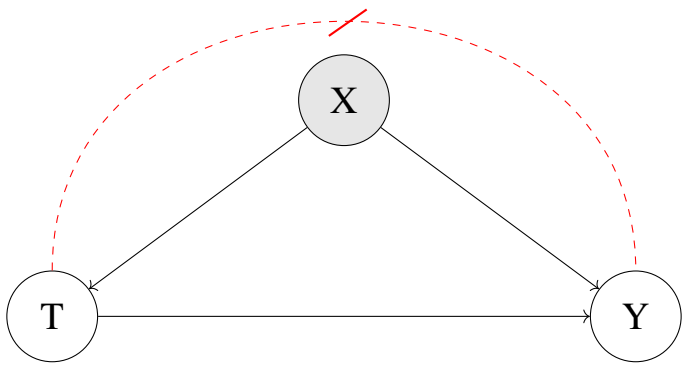

Figure 2.2: A causal DAG illustrating the blocked path when conditioning on the confounder X.

The presence of X as a common cause of both T and Y introduces confounding. If we do not account for X, the observed association between T and Y may be biased because X affects both.

To address confounding, we must control for the confounder X by conditioning on X. By doing so, we can isolate the causal effect of T on Y. Conditioning on the confounder X resolves the problem of confounding because it allows us to compare groups that are similar with respect to X, thereby eliminating the confounding influence that X has on both the treatment T and the outcome Y.

In the causal DAG, controlling for X conceptually "blocks" the backdoor path from T to Y through X, ensuring that the association between T and Y reflects the causal effect. This is illustrated in Figure 2.2, where the path from T to Y through X is blocked by conditioning on

X.

Here is an example of confounding: Consider a study examining the effect of exercise (T) on heart health (Y), with age (X) as a potential confounder. Older individuals might be less likely to exercise and more prone to poor heart health. Without accounting for age, the observed relationship between exercise and heart health could be misleading, as it may actually reflect the influence of age rather than exercise. By controlling for age, we can more accurately estimate the true effect of exercise on heart health.

By addressing confounding through conditioning on relevant variables, we ensure that the treatment groups are comparable. This brings us to the concept of conditional exchangeability, which further refines our ability to draw causal inferences from observational data.

Conditional Exchangeability

The final assumption necessary for identifying the CATE from observational data is conditional exchangeability. This assumption refines the concept of exchangeability, which posits that treatment assignment is independent of potential outcomes. However, in practice, simple exchangeability is often violated due to confounding variables that influence both the treatment and the outcome.

For instance, imagine a study evaluating the effect of an exercise program (treatment) on weight loss (outcome). Suppose more motivated individuals, who are likely to lose weight regardless of participation in the program, are also more likely to join the exercise program. Here, the treatment assignment (joining the exercise program) is not independent of the potential outcomes (weight loss). This situation violates the exchangeability assumption because those who receive the treatment (exercise) differ inherently from those who do not, potentially leading to biased estimates of the treatment effect.

To address this limitation, we refine the assumption using the concept of conditional exchangeability. This assumption posits that the treatment assignment is independent of the potential outcomes, given a set of observed covariates $\mathbf{X}$. Formally, this is expressed as:

$$(Y^1, Y^0) \perp\!\!\!\perp T \mid \mathbf{X}.$$

Conditional exchangeability allows us to equate the expected potential outcomes under control and treatment within each level of the covariates $\mathbf{X}$ to the corresponding observed outcomes within the control and treatment groups, respectively. Specifically, we have:

$$E[Y^1 \mid \mathbf{X}] = E[Y^1 \mid T = 1, \mathbf{X}]$$

and

$$E[Y^0 \mid \mathbf{X}] = E[Y^0 \mid T = 0, \mathbf{X}].$$

This refined assumption is crucial for accurately estimating causal effects from observational data, as it ensures that any differences in outcomes between treated and control groups can be attributed to the treatment itself, rather than confounding factors.

As a result of all these assumptions, the CATE can be expressed as:

$$\begin{aligned}
\tau(\mathbf{x}) &= E[Y^1 - Y^0 \mid \mathbf{X} = \mathbf{x}] \\
&= E[Y^1 \mid T = 1, \mathbf{X} = \mathbf{x}] - E[Y^0 \mid T = 0, \mathbf{X} = \mathbf{x}] \\
&= E[Y \mid T = 1, \mathbf{X} = \mathbf{x}] - E[Y \mid T = 0, \mathbf{X} = \mathbf{x}],
\end{aligned}$$

where the second equality uses the conditional exchangeability assumption, and the third equality applies the consistency assumption.

In summary, the framework established in this chapter provides a solid foundation for accurately identifying and estimating causal effects. With this theoretical groundwork in place, we now shift our focus to the practical challenge of estimating the CATE from data. In the next chapter, we will explore the methods employed to estimate the CATE, applying the principles and assumptions discussed here to real-world data scenarios.

Chapter 3

Methodology

In this chapter, we explore the methodologies employed for the estimation of the CATE. Traditional methods for estimating CATEs often rely on parametric models, such as linear or generalized linear models, due to their simplicity and ease of use. However, these models may fall short in capturing the true complexities of treatment effects, especially when the relationships between the covariates, treatment and outcome are complex. Such limitations can lead to significant bias, particularly if the model is misspecified.

Given these challenges, there has been a growing shift toward using more flexible machine learning techniques to estimate CATEs. The meta-learner framework, in particular, has gained prominence for its ability to leverage a wide range of machine learning algorithms. Meta-learners are versatile tools that are not confined to any specific machine learning method, allowing for a more tailored and effective approach to estimating treatment effects.

This chapter will detail the meta-learners and cross-fitting techniques that form the core of our analysis. Additionally, we will outline the simulation design used to evaluate these methods. Our objective is to understand how these different approaches perform across various settings, helping to identify the most appropriate learner for specific situations and to assess the necessity of applying cross-fitting. By employing these techniques, we aim to enhance the accuracy and reliability of the causal inferences derived from the data.

3.1 Meta-Learners

Estimating the CATE is an important task in causal inference, especially when dealing with complex datasets where the relationships between covariates, treatment, and outcomes may not be straightforward. While using a single supervised learning model to estimate the CATE is an option, this approach often encounters challenges, such as model misspecification and an

inability to capture nuanced interactions between treatment and covariates.

Meta-learners for estimating the CATE, as defined by Künzel et al. [3], are a class of algorithms that combine supervised learning methods (i.e., base learners) in a structured manner, while allowing flexibility in the choice of these base learners. Unlike a single model, meta-learners can adapt to various types of data and integrate seamlessly into existing data analysis pipelines, making them versatile tools in causal inference.

The meta-learners considered in this study operate by first using base learners to estimate what are known as nuisance functions. These functions, while not of immediate interest, are used for to estimate the CATE. Examples include the propensity score function $e(x)$ and various response functions such as $\mu(\mathbf{x})$, $\mu_0(\mathbf{x})$, $\mu_1(\mathbf{x})$, and $\mu(\mathbf{x}, t)$.

Once these nuisance functions are estimated, they are used to predict the pseudo-outcome ψ, which can be seen as an approximation of the CATE. The final step involves using a base learner to regress the pseudo-outcomes on the covariates to obtain the final CATE estimate. This step serves as a smoothing mechanism, mitigating the impact of potential extreme values in the pseudo-outcomes.

The base learners used for estimating the nuisance functions can vary and may differ from those used to model the pseudo-outcomes. This flexibility allows meta-learners to be tailored to the specific structure of the data, making them particularly effective in scenarios where traditional models may struggle. The choice of base learners could include methods such as Random Forest, Bayesian Additive Regression Trees (BART), Lasso, XGBoost, or Neural Networks.

In the following sections, we will explore and compare several meta-learning approaches, each distinguished by its unique methodology.

3.1.1 S-Learner

The S-learner, also known as g-computation, is a straightforward approach to estimating the CATE. The ”S” in S-learner stands for ”single,” reflecting the fact that this method relies on a single model to predict the pseudo-outcome ψ. The procedure for estimating the CATE is detailed in Algorithm 1.

Algorithm 1 S-learner

Input: Data: $D = \{(\mathbf{X}_i, Y_i, T_i) \mid i = 1, \cdots, N\}$
Output: CATE: $\hat{\tau}(\mathbf{x})$
Estimate: $\mu(\mathbf{x}, t) = E(Y_i \mid \mathbf{X}_i = \mathbf{x},\, T_i = t)$ in D
Predict: $\psi_i = \hat{\mu}(\mathbf{X}_i, 1) - \hat{\mu}(\mathbf{X}_i, 0)$ in D
Estimate: $\phi(\mathbf{x}) = E(\psi_i \mid \mathbf{X}_i = \mathbf{x})$ in D
Define: $\hat{\tau}(\mathbf{x}) = \hat{\phi}(\mathbf{x})$

The S-learner requires the estimation of only a single nuisance function, $\mu(\mathbf{x}, t)$, which can be effective if accurately estimated. This method is particularly suitable when the CATE is close to zero, as $\mu(\mathbf{x}, t)$ would likely be independent of the treatment t. However, this approach may introduce bias toward zero in the CATE, especially when the treatment effect is subtle or when a large number of covariates complicates the isolation of the treatment effect using a single model.

In our work, we assume that all relevant confounding variables are available and can be used for prediction. However, it is crucial to recognize that, in some scenarios, legal, ethical, or practical constraints may prevent the use of certain attributes in the prediction process. This situation, referred to as "runtime confounding" by Coston et al. [11], can occur in specific contexts. For instance, in child welfare screening, call workers decide which reports of abuse should be investigated based on call information and historical administrative data. If race implicitly influenced past decisions, it might not be appropriate to include race as a predictor in a model, despite its impact on outcomes.

3.1.2 T-Learner

The T-learner is introduced as another approach to estimate the CATE. The "T" in T-learner stands for "two," indicating the method's use of two separate models to estimate the pseudo-outcome: one for the treatment group and one for the control group. The estimation procedure is depicted in Algorithm 2.

Algorithm 2 T-learner

Input: Data: $D = \{(\mathbf{X}_i, Y_i, T_i) \mid i = 1, \cdots, N\}$
Output: CATE: $\hat{\tau}(\mathbf{x})$
Estimate: $\mu_1(\mathbf{x}) = E(Y_i \mid \mathbf{X}_i = \mathbf{x},\, T_i = 1)$ in D
Estimate: $\mu_0(\mathbf{x}) = E(Y_i \mid \mathbf{X}_i = \mathbf{x},\, T_i = 0)$ in D
Predict: $\psi_i = \hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i)$ in D
Estimate: $\phi(\mathbf{x}) = E(\psi_i \mid \mathbf{X}_i = \mathbf{x})$ in D
Define: $\hat{\tau}(\mathbf{x}) = \hat{\phi}(\mathbf{x})$

The T-learner offers a significant advantage over the S-learner by allowing separate models to capture the potentially different distributions of covariates in the treated and control groups. This flexibility can lead to more accurate estimates of the treatment effect, particularly when the CATE is complex, and the relationship between covariates and outcomes varies significantly between the two groups.

However, the T-learner has its limitations. One key drawback is the potential loss of data efficiency since the data is divided into treatment and control groups. This division can increase variance and bias in the CATE estimates, especially when sample sizes are small.

As noted by Morzywolek et al. [12], another limitation is that the T-learner does not directly target the CATE. Instead, it involves solving two separate minimization problems to generate outcome predictions for each treatment group. While these predictions may balance bias and variance within each outcome regression, this does not guarantee an optimal bias-variance trade-off for estimating the CATE when considering the difference between these predictions.

Thus far, the learners we've discussed do not incorporate the propensity score. The following learner, however, does utilize the propensity score in its approach.

3.1.3 IPW-Learner

We now introduce the Inverse Probability Weighting (IPW) learner, a method that leverages the propensity score to estimate the CATE. The IPW learner is designed to adjust for the fact that treatment assignment in observational studies is often not random. By weighting each observation according to the inverse of the probability of receiving the treatment they actually received, the IPW learner effectively balances the treatment and control groups concerning their covariate distributions. The key insight behind the IPW learner is to create a pseudo-population where the covariates are balanced between the treated and control groups, mimicking a randomized experiment. The workings of the IPW learner is depicted in Algorithm 3.

Algorithm 3 IPW-learner

Input: Data: $D = \{(\mathbf{X}_i, Y_i, T_i) \mid i = 1, \cdots, N\}$
Output: CATE: $\hat{\tau}(\mathbf{x})$
Estimate: $e(\mathbf{x}) = P(T_i = 1 \mid \mathbf{X}_i = \mathbf{x})$ in D
Predict: $\psi_i = Y_i(\frac{T_i}{\hat{e}(\mathbf{X}_i)} - \frac{1-T_i}{1-\hat{e}(\mathbf{X}_i)})$ in D
Estimate: $\phi(\mathbf{x}) = E(\psi_i \mid \mathbf{X}_i = \mathbf{x})$ in D
Define: $\hat{\tau}(\mathbf{x}) = \hat{\phi}(\mathbf{x})$

The IPW learner primarily depends on the accurate estimation of the propensity score. To prevent issues such as division by zero, we trim the estimated propensity scores $\hat{e}(\mathbf{x})$ to ensure they fall between 0.025 and 0.975. This trimming process is applied consistently across all learners that involve the estimation of propensity scores. The IPW learner is expected to perform well when the propensity score is relatively easy to estimate, typically when it is not strongly dependent on the covariates. However, performance may decline when the propensity score is highly dependent on covariates. Additionally, when the propensity score is close to 0 or 1 for certain observations, the corresponding weights can become excessively large, increasing the variance of the estimates and potentially leading to instability in the results.

3.1.4 DR-Learner

The DR-learner (Doubly Robust Learner) is a powerful method that combines the strengths of the T-learner and the IPW-learner to estimate the CATE. The core idea behind the DR-learner

is to leverage both outcome modeling, as in the T-learner, and propensity score weighting, as used in the IPW-learner. By integrating these two approaches, the DR-learner provides a doubly robust estimate, meaning that errors in the estimated propensity scores can be offset by more accurate outcome estimates, and vice versa.

In addition to its double robustness, the DR-learner benefits from what is known as the Neyman orthogonality property. This property ensures that small errors in estimating nuisance functions do not introduce significant bias in the treatment effect estimate, thereby enhancing the reliability of the CATE estimation.

The workings of the DR-learner are detailed in Algorithm 4.

Algorithm 4 DR-learner

Input: Data: $D = \{(\mathbf{X}_i, Y_i, T_i) \mid i = 1, \cdots, N\}$
Output: CATE: $\hat{\tau}(\mathbf{x})$
Estimate: $\mu_1(\mathbf{x}) = E(Y_i \mid \mathbf{X}_i = \mathbf{x},\, T_i = 1)$ in $D_{T_i=1}$
Estimate: $\mu_0(\mathbf{x}) = E(Y_i \mid \mathbf{X}_i = \mathbf{x},\, T_i = 0)$ in $D_{T_i=0}$
Estimate: $e(\mathbf{x}) = P(T_i = 1 \mid \mathbf{X}_i = \mathbf{x})$ in D
Predict: $\psi_i = \frac{T_i(Y_i - \hat{\mu}_1(\mathbf{X}_i))}{\hat{e}(\mathbf{X}_i)} - \frac{(1-T_i)(Y_i - \hat{\mu}_0(\mathbf{X}_i))}{1-\hat{e}(\mathbf{X}_i)} + \hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i)$ in D
Estimate: $\phi(\mathbf{x}) = E(\psi_i \mid \mathbf{X}_i = \mathbf{x})$ in D
Define: $\hat{\tau}(\mathbf{x}) = \hat{\phi}(\mathbf{x})$

The DR-learner is expected to perform well even in high-dimensional settings where either the propensity score model or the outcome model may be misspecified. By incorporating both models, the DR-learner can leverage the accurate specification of one model to mitigate the effects of the other's misspecification, resulting in more robust and stable CATE estimates. Additionally, the DR-learner tends to have lower variance compared to the IPW-learner, particularly when propensity scores are close to 0 or 1. This reduced variance arises because the DR-learner incorporates residuals from the outcome models, which help stabilize estimates and minimize the influence of extreme weights.

3.1.5 X-Learner

The X-learner is a versatile and robust method for estimating the CATE. Like the DR-learner, it utilizes the same nuisance functions but employs them differently to achieve its estimates. The process by which the X-learner operates is detailed in Algorithm 5.

Unlike other methods, the X-learner utilizes two pseudo-outcomes—one for the treated individuals and one for the control individuals—and then combines them using the propensity score. Specifically, it uses the treatment group data to predict individual treatment effects for the control group, and vice versa. These two pseudo-outcomes are then combined using the propensity

Algorithm 5 X-learner

Input: Data: $D = \{(\mathbf{X}_i, Y_i, T_i) \mid i = 1, \cdots, N\}$
Output: CATE: $\hat{\tau}(\mathbf{x})$
Estimate: $e(\mathbf{x}) = P(T_i = 1|\mathbf{X}_i = \mathbf{x})$ in D
Estimate: $\mu_1(\mathbf{x}) = E(Y_i \mid \mathbf{X}_i = \mathbf{x},\, T_i = 1)$ in $D_{T_i=1}$
Estimate: $\mu_0(\mathbf{x}) = E(Y_i \mid \mathbf{X}_i = \mathbf{x},\, T_i = 0)$ in $D_{T_i=0}$
Predict: $\psi_i^1 = Y_i - \hat{\mu}_0(\mathbf{X}_i)$ in $D_{T_i=1}$
Predict: $\psi_i^0 = \hat{\mu}_1(\mathbf{X}_i) - Y_i$ in $D_{T_i=0}$
Estimate: $\phi^1(\mathbf{x}) = E(\psi_i^1 \mid \mathbf{X}_i = \mathbf{x})$ in $D_{T_i=1}$
Estimate: $\phi^0(\mathbf{x}) = E(\psi_i^2 \mid \mathbf{X}_i = \mathbf{x})$ in $D_{T_i=0}$
Define: $\hat{\tau}(\mathbf{x}) = (1 - \hat{e}(\mathbf{x})) \cdot \hat{\phi}^1(\mathbf{x}) + \hat{e}(\mathbf{x}) \cdot \hat{\phi}^0(\mathbf{x})$

score. Künzel et al. [3] provided empirical evidence in their simulations that balancing the pseudo-outcomes with the propensity score performs better than both the S- and T-learners.

The X-learner is more data-efficient than the T-learner because it leverages information from both treated and control groups, reducing variance by utilizing all available data more effectively. Additionally, the use of the propensity score in the X-learner differs from its use in the DR-learner, meaning that issues related to extreme propensity scores are less likely to affect the X-learner.

However, the X-learner also has some drawbacks. Its accuracy is heavily dependent on the correct specification of both the outcome models and the propensity score. If either model is misspecified, the resulting CATE estimates may be biased. Additionally, the X-learner requires the estimation of three models for the nuisance functions and two additional models for the pseudo-outcomes. This complexity can increase the likelihood of producing a misspecified CATE. Moreover, just like the T-learner, the X-learner does not directly target the CATE.

3.1.6 R-Learner

The R-Learner, where ”R” stands for residualization, is an orthogonal learner designed to estimate the CATE with robustness to errors in nuisance parameter estimation. Residualization involves subtracting the predicted values of the outcome and treatment, based on covariates, to create residuals that capture the unexplained variation. The R-Learner then leverages these residuals to isolate the true treatment effect, $\tau(\mathbf{x})$. Like the DR-Learner, the R-Learner benefits from the orthogonality property, ensuring that small errors in estimating nuisance functions, such as the outcome model $\mu(\mathbf{x})$ and the propensity score $e(\mathbf{x})$, do not introduce significant bias into the treatment effect estimate. The detailed workings of the R-Learner are illustrated in Algorithm 6.

Due to the potential for high values in propensity scores, the pseudo-outcomes in the R-Learner are estimated by incorporating the squared residualized treatment as weights. This step is cru-

Algorithm 6 R-learner

Input: Data: $D = \{(\mathbf{X}_i, Y_i, T_i) \mid i = 1, \cdots, N\}$
Output: CATE: $\hat{\tau}(\mathbf{x})$
Estimate: $\mu(\mathbf{x}) = E(Y_i \mid \mathbf{X}_i = \mathbf{x})$ in D
Estimate: $e(\mathbf{x}) = P(T_i = 1 \mid \mathbf{X}_i = \mathbf{x})$ in D
Predict: $\psi_i = \frac{Y_i - \hat{\mu}(\mathbf{X}_i)}{T_i - \hat{e}(\mathbf{X}_i)}$ in D
Predict: $w_i = (T_i - \hat{e}(\mathbf{X}_i))^2$ in D
Estimate: $\phi(\mathbf{x}) = E(\psi_i \mid \mathbf{X}_i = \mathbf{x})$ with weights w_i in D
Define: $\hat{\tau}(\mathbf{x}) = \hat{\phi}(\mathbf{x})$

cial, as it ensures the R-Learner's orthogonality. Without it, the R-Learner would face similar issues as the IPW-Learner. One advantage of the R-Learner over the DR-Learner is its ability to mitigate the problem of extreme weights by effectively eliminating them. Additionally, the R-Learner's reliance on fewer models can lead to more accurate estimates, potentially making it more favorable than the X-Learner and DR-Learner in certain situations. Thanks to its orthogonal property, the R-Learner is expected to perform well even if the outcome model or the propensity score is not perfectly estimated.

3.2 Sample Splitting and Cross-Fitting

When estimating the CATE using meta-learners, it is crucial to ensure the accuracy and reliability of the estimates. The complexity of these methods requires careful attention to potential biases, particularly those stemming from overfitting.

To address these concerns, techniques such as sample splitting and cross-fitting have been developed. These methods help mitigate overfitting bias by partitioning the data in a way that enhances the reliability of causal effect estimates. Following the terminology of Okasa [7], we will delve deeper into these concepts.

Sample splitting is a technique used in causal effect estimation, particularly when multiple estimation steps are required. This method addresses the overfitting bias that can occur when the same data sample is used both to estimate nuisance functions and to estimate the causal effect. The process involves dividing the data into distinct parts: one part is used for estimating nuisance functions, while the other part is used for causal effect estimation. This approach controls overfitting bias but can increase the variance of the estimator due to the reduced data available for each step.

Double sample splitting is an advanced method that reduces the variance introduced by standard sample splitting while maintaining low bias. In this approach, each nuisance function is estimated on a separate portion of the data, rather than using a single split for all nuisance functions. The training data is divided into $M + 1$ equally sized parts, where M is the number of

nuisance functions to be estimated. Each of the M parts is used to estimate a different nuisance function, with the remaining part reserved for the final causal effect estimation.

To further reduce variance, cross-fitting switches the roles of the data parts so that each part is used for both nuisance function estimation and causal effect estimation in different iterations. The final causal effect estimator is then computed as the median or mean of the separate effect estimates produced in each iteration. This process can be extended to more than $M + 1$ splits, known as K-fold cross-fitting. This method aims to reduce the variance of the estimator while retaining the bias-reducing benefits of sample splitting, leading to more robust and reliable causal effect estimates.

We aim to apply cross-fitting uniformly across various meta-learners to identify the conditions under which different methods perform best. As discussed by Jacob [6], there are several approaches to implementing cross-fitting. In this section, we outline the approach applied specifically to the DR-learner. Similar procedures are used for creating the cross-fitting versions of the other learners. The workings of the cross-fitting algorithm we will use are depicted in Algorithm 7.

Algorithm 7 Cross-fitting

Input: Data: $D = \{(\mathbf{X}_i, Y_i, T_i) \mid i = 1, \cdots, N\}$, Folds: K, Option a
Output: CATE $\hat{\tau}(\mathbf{x})$
  Split the data in K equal folds: $I_1, \cdots, I_K$ such that $\bigcup_{k=1}^{K} I_k = \{1, \cdots, N\}$
  Define: $D_k = \{(\mathbf{X}_i, Y_i, T_i) \mid i \in I_k\}$

  for $k = 1, \cdots, K$ **do**:
    Define: $D_{\text{train1}} = D_{k+1 \bmod K}$, $D_{\text{train2}} = D_{k+2 \bmod K}$, $D_{\text{test}} = D_k$
    Estimate: $\mu_1(\mathbf{x}) = E(Y_i \mid \mathbf{X}_i = \mathbf{x},\, T_i = 1)$ in $D_{\text{train1},T_i=1}$
    Estimate: $\mu_0(\mathbf{x}) = E(Y_i \mid \mathbf{X}_i = \mathbf{x},\, T_i = 0)$ in $D_{\text{train1},T_i=0}$
    Estimate: $e(\mathbf{x}) = P(T_i = 1 \mid \mathbf{X}_i = \mathbf{x})$ in D_{train2}
    Predict: $\psi_i = \frac{T_i(Y_i - \hat{\mu}_1(\mathbf{X}_i))}{\hat{e}(\mathbf{X}_i)} - \frac{(1-T_i)(Y_i - \hat{\mu}_0(\mathbf{X}_i))}{1-\hat{e}(\mathbf{X}_i)} + \hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i)$ in D_{test}
    Estimate: $\phi(\mathbf{x}) = E(\psi_i \mid \mathbf{X}_i = \mathbf{x})$ in D_{test}
    Define: $\hat{\tau}_k(\mathbf{x}) = \hat{\phi}(\mathbf{x})$
  end for
  if $a =$ mean **then**
    Define: $\hat{\tau}(\mathbf{x}) = \text{mean}\{\hat{\tau}_1(\mathbf{x}), \cdots, \hat{\tau}_K(\mathbf{x})\}$
  else
    Define: $\hat{\tau}(\mathbf{x}) = \text{median}\{\hat{\tau}_1(\mathbf{x}), \cdots, \hat{\tau}_K(\mathbf{x})\}$
  end if

We begin by partitioning the dataset D, consisting of observations $(\mathbf{X}_i, Y_i, T_i)$ for $i = 1, \ldots, N$, into K approximately equal folds. Each fold is denoted as $I_1, I_2, \ldots, I_K$.

Next, the algorithm focuses on estimating the nuisance functions. For each fold k, the outcome model for the treatment group $\mu_1(\mathbf{x})$ is estimated using data from $D_{(k+1) \bmod K}$, and the outcome

model for the control group $\mu_0(\mathbf{x})$ is estimated using data from the same subset. The propensity score $e(\mathbf{x})$ is estimated using data from $D_{(k+2) \bmod K}$. This ensures that the nuisance functions are trained on different subsets of the data.

These trained nuisance functions are then applied to the data in D_k to predict the pseudo-outcome ψ. This pseudo-outcome is used to estimate the CATE within D_k, resulting in a k-fold CATE estimate. Finally, the overall CATE estimate can be computed as either the mean or the median of the k-fold estimates.

3.3 Simulation study

In this section, we describe a simulation study designed to evaluate the performance of the cross-fitting and meta-learner methods discussed in the previous sections. The primary goal is to assess the accuracy and robustness of these methods in estimating causal effects under various scenarios. We will use 6 full-sample meta-learners and apply a 5-fold median cross-fitting algorithms to each, totaling 12 different methods. This mix allows us to test a wide range of estimation approaches, from full-sample to cross-fitting. By trying out different algorithms, we can compare their performance and find the best fit for various scenarios.

3.3.1 Data Generation Process

In the process of data generation, we take into account several key parameters: the sample size n, the number of covariates p, and a value α in the range of 0 to 1 that controls the density (or sparsity) of the correlation matrix. Additionally, we consider the response function under control μ_0, the propensity score function e, and the CATE τ.

Covariate Generation

We generate the covariates $\mathbf{X} \in \mathbb{R}^{n \times p}$ based on the correlation matrix $\mathbf{\Sigma}(\alpha) \in \mathbb{R}^{p \times p}$, where α regulates the density of the correlation matrix. These correlation matrices are created using the `make_sparse_spd_matrix(norm_diag=True, dim=`p`, alpha=`α`)` function from the `sklearn.datasets` package. This function allows us to control the density level of the correlation matrix, which impacts the relationships between the covariates.

For the ith observation, the p covariates $\mathbf{x}_i$ are generated following a multivariate normal distribution with mean vector $\mathbf{0}$ and covariance matrix $\mathbf{\Sigma}(\alpha)$:

$$\mathbf{x}_i \overset{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}(\alpha)).$$

Potential Outcomes Generation

To generate the observed outcome $\mathbf{Y} \in \mathbb{R}^n$, we first construct the noise-free potential outcomes $\mathbf{M}^0$ and $\mathbf{M}^1 \in \mathbb{R}^n$, and the noisy potential outcomes $\mathbf{Y}^0$ and $\mathbf{Y}^1 \in \mathbb{R}^n$. This is done using the response function under control μ_0, the CATE τ, and error terms ε^0 and ε^1:

$$\begin{aligned} y_i^0 &= m_i^0 + \varepsilon_i^0, & m_i^0 &= \mu_0(\mathbf{x}_i), \quad \varepsilon_i^0 \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\varepsilon), \\ y_i^1 &= m_i^1 + \varepsilon_i^1, & m_i^1 &= \mu_0(\mathbf{x}_i) + \tau(\mathbf{x}_i), \quad \varepsilon_i^1 \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\varepsilon), \end{aligned}$$

where σ_ε is the standard deviation of the error terms ε_i^0 and ε_i^1, which introduce variability into the potential outcomes.

Treatment Assignment

Next, we generate the treatment variable $\mathbf{T} \in \{0, 1\}^n$ using the propensity score function e:

$$t_i \overset{\text{iid}}{\sim} \text{Bern}(e(\mathbf{x}_i)).$$

The treatment assignments t_i are drawn from a Bernoulli distribution with parameter $e(\mathbf{x}_i)$.

Observed Outcomes

With the potential outcomes and treatment assignments in place, we generate the observed outcome $\mathbf{Y}$ as follows:

$$y_i = t_i \cdot y_i^1 + (1 - t_i) \cdot y_i^0.$$

The observed outcome y_i is a combination of the noisy potential outcomes y_i^0 and y_i^1 based on the treatment assignment.

Observed Data Matrix

Finally, the observed data can be encapsulated in the matrix $\mathbf{D} \in \mathbb{R}^{n \times (p+2)}$, which includes the covariates, treatment assignments, and observed outcomes:

$$\mathbf{D} = (\mathbf{X}, \mathbf{T}, \mathbf{Y}).$$

This matrix serves as the basis for analyzing the effect of treatments on outcomes, using the generated data to estimate causal relationships.

3.3.2 Simulation Scenarios

In this section, we outline the different simulation scenarios used to evaluate the performance of the causal inference methods. We vary several parameters to create diverse conditions under which the methods are tested.

Baseline Scenario

The baseline scenario serves as a reference point for evaluating the performance of the meta-learners. The parameters for the baseline scenario are as follows:

We choose the sample size n to be 2000, ensuring a sufficient amount of data for reliable estimation and to allow for the evaluation of model performance under realistic conditions. The number of covariates p is set to 20, balancing complexity and practicality. The standard deviation for the error terms, σ_ε, is set to 1, a common choice in simulation studies to maintain a controlled environment for evaluating different methods.

To determine the appropriate value of α for generating the correlation matrix with a target density of 30%, we employed an iterative search approach. The density of a matrix is defined as the proportion of non-zero elements in the matrix. Mathematically, the density D of a matrix A can be calculated as:

$$D = \frac{\sum_{i=1}^{n} \sum_{j=1}^{m} \mathbb{I}(|a_{ij}| > 0)}{n \cdot m}$$

where n and m are the dimensions of the matrix A, a_{ij} represents the element in the i-th row and j-th column of A, and $\mathbb{I}(\cdot)$ is the indicator function. We generated matrices using the function `make_sparse_spd_matrix` with varying values of α. For each α in the range from 0 to 1 with increments of 0.01, we calculated the mean density over multiple iterations. The optimal α was then selected as the one that produced a mean density closest to the target density of 30%. The optimal parameter found through this method is $\alpha = 0.85$.

In our simulations, we use Random Forest (RF) as the base learner for the meta-learners. This choice is motivated by several key factors. First, RF offers a high degree of flexibility with numerous parameters, allowing us to fine-tune the model to balance between overfitting and underfitting, depending on the precision required in estimating the objective function. Additionally, Random Forest is versatile in handling both binary and continuous outcomes, making it well-suited for the diverse tasks involved in our simulations.

For our baseline scenario, we have carefully selected the initial settings for the Random Forest algorithm to balance model complexity and computational efficiency. The parameters for the baseline scenario are as follows:

- `n_trees = 500`: We set the number of trees to 500. This choice provides a good

balance between computational efficiency and model performance. With 500 trees, the Random Forest can capture a wide range of patterns in the data while avoiding excessive computational cost.

- `max_depth = None`: Allowing the trees to grow to their maximum depth. This parameter is set to `None` to ensure that each tree can grow until all leaves are pure or until all leaves contain less than the minimum number of samples required to split further. This setting allows the model to fully explore the complexity of the data.

- `min_samples_leaf = 5`: Setting the minimum number of samples required to be at a leaf node to 5. This constraint helps prevent the model from overfitting by ensuring that each leaf has a sufficient number of samples, thus making the model more generalizable to new data.

- `max_features = 'sqrt'`: Using the square root of the total number of features when looking for the best split. This setting helps to reduce correlation among trees, leading to a more robust model. By considering a subset of features at each split, the Random Forest can better generalize and avoid overfitting to the training data.

These baseline settings are chosen to provide a robust starting point for evaluating the performance of the Random Forest algorithm. By balancing model complexity and computational efficiency, we can ensure that the baseline scenario offers a reliable reference for comparison with other configurations. The combination of these parameters allows the Random Forest to capture the underlying patterns in the data effectively while maintaining generalizability and avoiding overfitting.

For the CATE, we selected a simple 2-dimensional function that depends on two covariates x_1 and x_2. The chosen CATE function is:

$$\tau(\mathbf{x}) = 1 + 0.5x_1 + 0.3x_2$$

The rationale for selecting this CATE function is its simplicity, making it an excellent baseline for our study.

For the choice of the response function and propensity scores, we have drawn inspiration from [13] but with a few modifications. First, we define the propensity score function $e(\mathbf{x}; c_{1e}, c_{2e})$ as:

$$e(\mathbf{x}; c_{1e}, c_{2e}) = \frac{1}{1 + \exp\left(c_{1e}x_1 + \frac{c_{2e}}{1+\exp(-x_3)}\right)}$$

and the response function $\mu_0(\mathbf{x}; c_{1\mu}, c_{2\mu})$ as:

$$\mu_0(\mathbf{x}; c_{1\mu}, c_{2\mu}) = \frac{1}{1 + \exp(-c_{1\mu} x_1)} + c_{2\mu} x_3.$$

To find the coefficients for the response function μ_0 and the propensity score function e, we use the coefficient of determination, R^2. The R^2 value measures the proportion of the variance in the dependent variable that is predictable from the independent variables. Mathematically, R^2 is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where y_i are the observed values, $\hat{y}_i$ are the predicted values, and $\bar{y}$ is the mean of the observed values. An R^2 value of 0 indicates that the model does not explain any of the variance, while an R^2 value of 1 indicates that the model explains all the variance.

For each combination of coefficients c_{1e}, c_{2e}, $c_{1\mu}$, and $c_{2\mu}$, we generate data $\mathbf{D} = (\mathbf{X}, Y, T)$. We fit a logistic regression model to predict treatment assignments T from the covariates $\mathbf{X}$ and calculate the R^2 value to measure how well the covariates explain the variance in the treatment assignments. Here, the R^2 is actually Efron's pseudo R^2, which takes a form similar to the one defined earlier:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(t_i - \hat{e}(\mathbf{x}_i))^2}{\sum_{i=1}^{n}(t_i - \bar{t})^2},$$

where $\hat{e}(\mathbf{x}_i)$ represents the predicted value from the logistic regression.

We then fit a linear regression model for the control group and to predict the outcomes Y from the covariates $\mathbf{X}$ and calculate the R^2 value to measure how well the covariates explain the variance in the outcomes.

To find the parameters for the baseline scenario, we perform a grid search over a range of values for c_{1e}, c_{2e}, $c_{1\mu}$, and $c_{2\mu}$. For each combination, we calculate the R^2 values for the propensity score function R_e^2 and the outcome function R_μ^2. We then select the combination of c_{1e} and c_{2e} that results in an R_e^2 value close to a low target value (0.05 for very low confounding) and the combination of $c_{1\mu}$ and $c_{2\mu}$ that results in an R_μ^2 value close to 0.5. Using this method, we found the following parameters: $c_{1e} = 0.01$, $c_{2e} = 0.03775$, $c_{1\mu} = 2.25$, and $c_{2\mu} = 1$. We observe that c_{2e} is very small, indicating that the propensity score function is well-balanced when $x_1 = 0$ but becomes less balanced as x_1 deviates from 0.

Varying Sample Sizes

To evaluate the robustness of the meta-learners under different conditions, we vary the sample size n. This helps to understand how the performance of the methods scales with the amount of available data. We consider the following sample sizes:

- $n = 500$: A smaller sample size to assess the performance in limited data scenarios.
- $n = 5000$: A larger sample size to evaluate performance with abundant data.

By examining various sample sizes, we can assess the sensitivity of the meta-learners to the amount of data available and identify any potential issues related to overfitting or underfitting. We expect all learners, including their cross-fitting versions, to perform better as the sample size increases.

Varying Covariate Sizes

In addition to varying the sample size, we also examine the impact of different numbers of covariates p. This analysis helps us understand how the introduction of additional noise variables affects the performance of the meta-learners. We consider the following covariate sizes:

- $p = 5$: A small number of covariates to test the performance in simpler scenarios.
- $p = 50$: A larger number of covariates to evaluate the methods' handling of more complex data.

We expect that the estimation process is going to be more challenging for higher number of covariates.

Varying Noise Levels

To examine the impact of noise on the outcome, we vary the standard deviation of the error terms σ_ε. This allows us to assess the sensitivity of the methods to noise in the outcome. The values of σ_ε are selected based on R^2_μ, with R^2_μ set to 0.8 to represent low noise and 0.2 to represent high noise. This results in the following noise levels:

- $\sigma_\varepsilon = 0.45$: A small noise level to test performance in near-ideal conditions with minimal variability.
- $\sigma_\varepsilon = 2.07$: A large noise level to evaluate the robustness of the methods under substantial variability.

We expect all learners to perform better with lower noise levels compared to higher ones. However, the degree of improvement will likely vary, as each learner employs the response functions

differently.

Varying Levels of Density of the Correlation Matrix

We will also explore how different levels of density in the correlation matrix affect the performance of meta-learners. By varying the density, we can simulate different degrees of interdependencies between the covariates. We consider two scenarios:

- **Low Density:** We set the density of the correlation matrix to 0.15 which corresponds to an α of 0.92. This scenario represents a sparse correlation matrix where most elements are zero, indicating weak or no relationships among the majority of covariates.
- **High Density:** We set the density of the correlation matrix to 0.85, which corresponds to an α of 0.48. This scenario represents a dense correlation matrix where most elements are non-zero, indicating strong interdependencies among the covariates.

By comparing the results under these two scenarios, we can assess how the complexity and interdependencies among covariates influence the performance of the meta-learners. This analysis helps us understand the impact of covariate structure on the meta-learners.

Shallow Trees vs. Deep Trees

We compare the performance of Random Forests by introducing two configurations: shallow trees with more constraints and deep trees with less regularization. For both configurations, we keep the number of trees constant at 500.

Shallow Trees with More Constraints: This configuration aims to reduce overfitting by limiting the depth of the trees and increasing the minimum number of samples required to be at a leaf node. Additionally, fewer features are considered at each split.

The parameters for this setting are:

- `max_depth = 10`: Limits the depth of the trees, making them shallower.
- `min_samples_leaf = 10`: Increases the minimum samples per leaf, which helps generalize the trees.
- `max_features = 'log2'`: Uses fewer features at each split to reduce correlation between trees.

Deep Trees with Less Regularization: This configuration allows the trees to grow deeper and uses more features at each split. This setting might help in capturing more complex relationships in the data.

The parameters for this setting are:

- `max_depth = None`: Allows trees to grow to their maximum depth.
- `min_samples_leaf = 2`: Decreases the minimum samples per leaf, which makes the trees more flexible.
- `max_features = None`: Considers all features at each split to potentially capture more complex interactions.

By comparing these settings, we aim to understand the trade-offs between model complexity and regularization. Shallow trees are expected to generalize better and reduce overfitting, while deep trees might capture more intricate patterns at the risk of overfitting the training data.

Varying Levels of Confounding

In this subsection, we examine how different levels of confounding impact the performance of meta-learners. We simulate varying levels of confounding by adjusting the R_e^2 value for the propensity score function $e(x)$, while keeping the R_{mu}^2 value for the response function $\mu_0(x)$ constant. To explore the effect of confounding, we consider two scenarios:

- **Low Confounding:** We set the R_e^2 value for the propensity score function $e(x)$ to 0.2, corresponding to $c_{1e} = 1.15$ and $c_{2e} = 0.1$. This scenario represents a lower degree of confounding, where the treatment assignment is less predictable based on the covariates.
- **High Confounding:** We set the R_e^2 value for the propensity score function $e(x)$ to 0.8, corresponding to $c_{1e} = 7.525$ and $c_{2e} = 0.1$. This scenario represents a higher degree of confounding, where the treatment assignment is more strongly influenced by the covariates.

The R^2 value for the response function $\mu_0(x)$ is kept constant at 0.5 in both scenarios. This allows us to isolate the effect of varying levels of confounding on the performance of the meta-learners.

By comparing the results under these two scenarios, we can assess the sensitivity of the meta-learners to different levels of confounding and identify which methods are more robust in the presence of confounding.

Varying Balance of Propensity Scores

In addition to examining the impact of confounding, it is crucial to understand how the balance of the propensity score affects the performance of meta-learners. The propensity score $e(\mathbf{x})$ represents the probability of receiving treatment given the covariates. Balanced propensity

scores indicate a more even distribution of treated and control units, while unbalanced propensity scores suggest a skewed distribution, which can affect the reliability of causal estimates. To investigate the impact of balance on propensity scores, we use the following function:

$$e(\mathbf{x}; c_{1e}, c_{2e}) = \frac{1}{1 + \exp\left(c_{1e}x_1 + c_{2e}\right)},$$

which allows us to control the balance while maintaining the dependency on $\mathbf{x}$.

To assess the effects of balance, we modify the coefficients so that when $x_1 = 0$, we have $e(\mathbf{x}) = 0.5$ in the balanced case and $e(\mathbf{x}) = 0.05$ in the unbalanced case. We consider two scenarios:

- **Balanced Propensity Score**
  In this scenario, the probability of receiving treatment is more evenly distributed, corresponding to $c_{1e} = 0.35$ and $c_{2e} = 0$. This setup ensures a balanced distribution between treated and control groups.

- **Unbalanced Propensity Score**
  In this scenario, the probability of receiving treatment is much lower, leading to an unbalanced design where the majority of units belong to the control group. This setup corresponds to $c_{1e} = 0.66$ and $c_{2e} = \ln(19)$, and tests the robustness of the meta-learners in handling highly imbalanced data, which is common in real-world applications where the treatment group is significantly smaller than the control group.

By comparing the performance of the meta-learners under these two propensity score scenarios, we can evaluate their ability to handle different levels of balance in treatment assignment. This analysis will help identify which meta-learners are more robust to imbalances.

Complexity of Propensity Scores

We also aim to explore the impact of varying complexity in propensity score functions on the performance of meta-learners. Specifically, we compare a scenario where the propensity score is constant at 0.5 against a more complex scenario, where the propensity score follows the baseline scenario but with increased complexity by setting $c_{1e} = 0.5$ and $c_{2e} = 5$.

This comparison allows us to assess how the complexity of the propensity score function influences the accuracy and robustness of the meta-learners.

- **Constant Propensity Score**
  In this scenario, the propensity score is fixed at 0.5 for all observations, representing a simplified situation where the treatment assignment is completely random and indepen-

dent of covariates.

- **Complex Propensity Score**
  Here, we introduce complexity by modifying the coefficients in the baseline propensity score model to $c_{1e} = 0.5$ and $c_{2e} = 5$. This adjustment creates a more intricate relationship between the covariates and the treatment assignment, simulating real-world scenarios where the treatment is not randomly assigned and is instead influenced by specific covariate patterns.

By comparing the performance of the meta-learners in these two scenarios, we will gain insights into their ability to handle varying levels of complexity in propensity score functions. This analysis is crucial for understanding the robustness of these methods in real-world applications, where the treatment assignment mechanism is often complex.

Comparing CATE with Varying Levels of Covariate Dependence

To gain further insight into the performance of the meta-learners, we compare scenarios where the CATE depends on the amount of covariates. This analysis helps to understand how well the learners handle varying levels of complexity in the treatment effect function.

- **CATE Independent of Covariates:** $\tau(\mathbf{x}) = 1$
  In this scenario, the CATE is independent of covariates and is represented by a constant function. This setting tests the meta-learners' ability to identify and accurately estimate treatment effects that are straightforward and involve minimal complexity.

- **CATE Dependent on Multiple Covariates:** $\tau(\mathbf{x}) = 1 + \sum_{j=1}^{15} \frac{0.5}{j} x_j$
  Here, the CATE depends on 15 covariates, with the importance of each covariate diminishing as the coefficient decreases. This scenario introduces higher complexity, requiring the meta-learners to model multiple covariates and their contributions to the treatment effect. The formula ensures that while all 15 covariates are involved, their impact is weighted, creating a gradient of importance.

By comparing these two CATE scenarios, we aim to evaluate the meta-learners' ability to manage varying complexities in the treatment effect. This comparison will highlight the strengths and weaknesses of each learner in capturing both simple and intricate relationships between covariates and the treatment effect.

Comparing No Treatment Effect CATE with Complex CATE

In addition to varying levels of covariate dependence, it is crucial to compare scenarios where there is no treatment effect against scenarios with a complex treatment effect. This comparison will help us understand how well the meta-learners can differentiate between cases with no

effect and those with intricate relationships between covariates and treatment.

- **No Treatment Effect:** $\tau(\mathbf{x}) = 0$
  In this scenario, there is no treatment effect, meaning that the CATE is zero for all units. This situation tests the meta-learners' ability to correctly identify the absence of a treatment effect.

- **Complex Treatment Effect:** $\tau(\mathbf{x}) = \sin(\pi x_1 x_2) + 2 \cdot (x_3 - 0.5)^2 + x_4 + 0.5x_5$
  In this scenario, the CATE is a complex function that involves multiple covariates and non-linear interactions among them. This function, also utilized in Okasa [7], is designed to test the ability of the meta-learners to accurately capture the intricate and non-linear relationships between the covariates and the treatment effect.

By comparing these two scenarios—no treatment effect versus a complex treatment effect—we aim to evaluate the sensitivity and accuracy of the meta-learners in detecting true treatment effects. This analysis will provide insights into which learners are more robust in scenarios with varying levels of treatment effect complexity.

Cross-Fitting Settings

We explore two different cross-fitting settings:

- **Cross-Fitting with $K = 3$:** In this scenario, we perform cross-fitting with $K = 3$ folds. This setting allows us to assess the performance of the meta-learners when using a smaller number of folds, which may offer a trade-off between computational efficiency and bias reduction.

- **Cross-Fitting with $K = 5$ Using the Mean:** In this scenario, we perform cross-fitting with $K = 5$ folds but instead of using the median, we average the k-fold estimates using the mean. This approach helps us understand the impact of using the mean on the overall effectiveness of cross-fitting, particularly in terms of reducing bias and variance.

By considering these different settings, we seek to gain insight into how the choice of cross-fitting parameters affects the performance of different meta-learners.

3.3.3 Performance Measures

To evaluate the performance of the algorithms, we reserve 10,000 observations from each setup for validation. This separate dataset allows us to assess how well our methods generalize to new, unseen data, providing a reliable measure of their effectiveness beyond the training data.

To ensure robustness in our evaluations, we repeat the estimation process 100 times for each

setup. This repetition helps us capture the variability in our results and assess the stability and reliability of our methods.

We consider three key metrics for validation: average mean squared error, average absolute bias, and average variance. These metrics provide a comprehensive assessment of the accuracy and reliability of the estimated CATE predictions. The evaluation procedure follows the methodology outlined by Jacob [6].

Average Mean Squared Error

To evaluate the accuracy of the CATE estimates, we utilize the Mean Squared Error (MSE) to construct a metric that we call the average MSE. This metric captures the overall accuracy of the estimations across all observations and is calculated in two steps:

First, for each individual observation i, the MSE is calculated based on the R CATE estimates $\hat{\tau}_r(\mathbf{X}_i)$ and the true CATE $\tau(\mathbf{X}_i)$:

$$MSE_i = \frac{1}{R} \sum_{r=1}^{R} \left[\hat{\tau}_r(\mathbf{X}_i) - \tau(\mathbf{X}_i)\right]^2 .$$

Next, the average MSE is determined by averaging the individual MSEs over all test data observations (N_T):

$$\overline{MSE} = \frac{1}{N_T} \sum_{i=1}^{N_T} MSE_i.$$

Average Absolute Bias

The average absolute bias uses absolute difference between the mean estimated CATE, $\overline{\hat{\tau}}(\mathbf{X}_i) = \frac{1}{R}\sum_{r=1}^{R} \hat{\tau}_r(\mathbf{X}_i)$, and the true CATE, $\tau(\mathbf{X}_i)$. This metric is crucial for detecting any systematic overestimation or underestimation in the predictions. The calculation is performed in two steps:

First, for each observation i, the absolute bias is computed:

$$|Bias_i| = \left|\overline{\hat{\tau}}(\mathbf{X}_i) - \tau(\mathbf{X}_i)\right| .$$

Then, the overall average absolute bias is obtained by averaging the individual absolute biases across all test observations:

$$\overline{|Bias|} = \frac{1}{N_T} \sum_{i=1}^{N_T} |Bias_i|.$$

Average Variance

The average variance measures the variability of the estimated CATE across the R repetitions for each observation. This metric provides insight into the consistency and stability of the predictions. The calculation is performed in two steps:

First, for each observation i, the variance of the estimated CATE is computed over the R repetitions:

$$Var_i = \frac{1}{R} \sum_{r=1}^{R} \left[\hat{\tau}_r(\mathbf{X}_i) - \overline{\hat{\tau}}(\mathbf{X}_i) \right]^2 .$$

Then, the average variance is obtained by averaging the individual variances across all test observations:

$$\overline{Var} = \frac{1}{N_T} \sum_{i=1}^{N_T} Var_i.$$

As discussed before, we set the test data set N_T to 10 000 and the repetitions R to 100. This approach provides a comprehensive view of each meta-learner's performance, allowing us to compare the accuracy (MSE), bias, and stability (variance) of the CATE estimates effectively. By employing these performance measures, we ensure a thorough evaluation of the meta-learners, facilitating a clear understanding of their strengths and weaknesses in estimating the CATE.

In the next chapter, we will present and discuss the results of our simulation study. Moving forward, when we refer to MSE, bias, and variance, we are specifically talking about the performance measures outlined in this section.

Chapter 4

Results

This chapter presents the findings from an extensive simulation study designed to evaluate the performance of various meta-learners. The primary objective of the study was to assess the accuracy of these learners under different conditions, with a particular emphasis on the impact of cross-fitting on their estimation accuracy. By systematically varying key parameters, we aimed to uncover the strengths and limitations of each method in estimating the CATE.

The chapter begins with an analysis of the baseline scenario, which serves as a benchmark for comparing the performance of the meta-learners across various settings. This is followed by a detailed presentation and discussion of the results from the additional scenarios.

4.1 Baseline Scenario

The baseline scenario offers a controlled environment where the key parameters of the simulation are set to standard values. This scenario serves as a benchmark for evaluating the performance of the meta-learners when exposed to variations in data complexity, sample size, and other factors. The results of the baseline scenario are presented in table 4.1. The metrics we have used here are the ones discussed in the performance measures section of the previous chapter.

In the baseline scenario, all learners perform worse when cross-fitting is applied, which is somewhat unexpected and might be due to the low sample size. The smaller subsets used in 5-fold cross-fitting may result in each k-fold CATE estimator being less accurate, leading to an overall less accurate median. The IPW-learner, in particular, exhibits the poorest performance, primarily due to high variance—a common issue when dealing with extreme propensity scores close to 0 or 1. These scores can inflate pseudo-outcomes, increasing variability and contributing to a high MSE. Additionally, the IPW-learner's high bias further exacerbates the MSE. Interest-

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.2479	0.4124	0.0119	0.5411	0.6374	0.0133
T-Learner	0.0921	0.1954	0.0228	0.1491	0.2642	0.0330
IPW-Learner	0.3357	0.3818	0.1049	0.5338	0.2629	0.4238
DR-Learner	0.0933	0.1526	0.0480	0.2297	0.1828	0.1731
X-Learner	0.0861	0.1790	0.0286	0.1296	0.2396	0.0334
R-Learner	0.1094	0.1246	0.0783	0.2177	0.1962	0.1512

Table 4.1: Metrics for Baseline Scenario

ingly, in the cross-fitting version of the IPW-learner, despite the increase in variance, the bias decreases.

The S-learner performs poorly compared to other learners, primarily due to its high bias rather than variance. Its reliance on a single model to estimate $\mu(\mathbf{x}, t)$ becomes problematic with a large number of covariates, making it challenging to accurately capture the treatment effect. For example, if the true $\mu(\mathbf{x}, t)$ depends only on covariates x_1, x_2, and the treatment t, introducing irrelevant noise variables can cause the model to incorrectly allocate importance, diluting the influence of the relevant covariates and leading to a biased estimation of the treatment effect. This would explain the S-learner's low variance but high bias, indicating its struggle to detect the treatment effect accurately.

For the other learners, the metrics are relatively close to each other. The R-learner shows the least bias, followed closely by the DR-learner. This might be attributed to the fact that both are orthogonal learners, making them more robust against errors in the specification of the nuisance functions.

Finally, the T-learner and X-learner both exhibit lower variance but higher bias compared to the R- and DR- learners. The T-learner, despite its simplicity, performs well in estimating the CATE, likely due to the generally balanced distribution of covariates, although some imbalance occurs as x_1 deviates far from zero. The X-learner slightly outperforms the T-learner, possibly because its use of the propensity score helps mitigate this imbalance. Both learners show similar variances, suggesting that the additional models in the X-learner offset the advantage of using more data. Their corresponding cross-fitted versions perform worse than the full-sample versions, but this degradation occurs similarly, which is expected since the X-learner and T-learner are quite similar when the propensity score is balanced.

4.2 Impact of Sample Size, Covariate Dimensions, Correlation Density, and Outcome Noise

This section presents the results of our analysis under varying conditions, including different sample sizes, covariate space dimensions, correlation matrix densities, and levels of noise in the outcomes.

Learner	**Full Sample**			**Cross-Fitting**		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.5158	0.6232	0.0156	0.9035	0.8549	0.0100
T-Learner	0.1414	0.2475	0.0391	0.2567	0.3357	0.0778
IPW-Learner	0.4387	0.4222	0.1728	0.8331	0.3118	0.6790
DR-Learner	0.1578	0.2027	0.0865	0.4559	0.2414	0.3586
X-Learner	0.1346	0.2319	0.0448	0.2094	0.3034	0.0613
R-Learner	0.1915	0.1729	0.1385	0.3940	0.2549	0.2862

Table 4.2: Metrics for Low Sample Size

Learner	**Full Sample**			**Cross-Fitting**		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.1595	0.3203	0.0109	0.3376	0.4901	0.0110
T-Learner	0.0691	0.1673	0.0168	0.1102	0.2249	0.0233
IPW-Learner	0.2771	0.3565	0.0730	0.3933	0.2148	0.3151
DR-Learner	0.0646	0.1252	0.0333	0.1478	0.1401	0.1121
X-Learner	0.0631	0.1492	0.0215	0.0963	0.2013	0.0264
R-Learner	0.0734	0.0957	0.0543	0.1456	0.1472	0.1051

Table 4.3: Metrics for High Sample Size

The results of varying sample sizes are presented in Tables 4.2 and 4.3. As anticipated, increasing the sample size consistently improves the accuracy of all learners. Additionally, the differences in MSE between full sample and cross-fitting estimates diminish with larger sample sizes, as more data leads to more accurate fold estimates. The relative performance of the learners remains consistent with the baseline scenario. These findings align with those of Okasa [7], where larger sample sizes enhance cross-fitting performance. However, it is important to note that under certain circumstances, cross-fitting performs better than the full sample learner, even with a sample size of 2000, which is not observed in our results.

The results of varying covariate dimensions are presented in Tables 4.4 and 4.5. As the covariate dimension increases, we observe a consistent trend: the bias of the estimates increases while the variance decreases. THis is usually what we see when we have underfitting. This may be attributed to the increased complexity of accurately estimating the nuisance functions with more covariates, as previously discussed.

These shifts in bias and variance result in the X-learner and T-learner being the top performers

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.1018	0.1938	0.0316	0.2899	0.4072	0.0356
T-Learner	0.0774	0.1044	0.0473	0.1675	0.2160	0.0735
IPW-Learner	0.4161	0.3219	0.2476	1.2209	0.1160	1.1940
DR-Learner	0.1194	0.0649	0.1070	0.3303	0.0785	0.3151
X-Learner	0.0748	0.0867	0.0534	0.1177	0.1623	0.0589
R-Learner	0.1726	0.0515	0.1664	0.2744	0.0860	0.2543

Table 4.4: Metrics for Low Dimensional Covariate Space

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.5463	0.6129	0.0073	0.9101	0.8137	0.0073
T-Learner	0.2045	0.3293	0.0176	0.2986	0.4076	0.0288
IPW-Learner	0.4491	0.4898	0.0690	0.5359	0.4024	0.2793
DR-Learner	0.1787	0.2853	0.0354	0.3031	0.3261	0.1289
X-Learner	0.1949	0.3202	0.0216	0.2603	0.3768	0.0278
R-Learner	0.1725	0.2518	0.0599	0.3002	0.3343	0.1148

Table 4.5: Metrics for High Dimensional Covariate Space

in low-dimensional settings, while the R-learner and DR-learner excel in high-dimensional scenarios. The S-learner's performance, however, deteriorates from being the third-best performer to the worst. This decline is likely due to the increased bias as the number of covariates rises, complicating the accurate estimation of $\mu(\mathbf{x}, t)$.

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.3111	0.4463	0.0101	0.6723	0.6813	0.0144
T-Learner	0.1436	0.2546	0.0214	0.2519	0.3575	0.0360
IPW-Learner	0.3988	0.4274	0.1057	0.6154	0.3284	0.4409
DR-Learner	0.1208	0.1922	0.0474	0.2705	0.2382	0.1716
X-Learner	0.1307	0.2353	0.0272	0.2148	0.3259	0.0343
R-Learner	0.1240	0.1484	0.0791	0.2586	0.2444	0.1533

Table 4.6: Metrics for low Correlation

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.3842	0.4942	0.0149	0.7581	0.7213	0.0170
T-Learner	0.1672	0.2770	0.0260	0.2573	0.3592	0.0366
IPW-Learner	0.4083	0.4336	0.1024	0.6099	0.3240	0.4383
DR-Learner	0.1482	0.2285	0.0511	0.2848	0.2536	0.1727
X-Learner	0.1580	0.2623	0.0333	0.2276	0.3327	0.0380
R-Learner	0.1492	0.1925	0.0813	0.2725	0.2587	0.1547

Table 4.7: Metrics for High Correlation

The results for varying the density of the correlation matrix are presented in Tables 4.6 and 4.7. Increasing the correlation between covariates generally leads to a decline in the performance of the learners in terms of both bias and variance. This decline may be attributed to high correlation introducing redundancy, which effectively reduces the amount of independent information available and diminishes the effectiveness of the learning process. Despite this, the relative performance differences among the learners remain consistent with those observed in the baseline scenario, indicating that while correlation affects all learners, their inherent strengths and weaknesses are preserved.

Learner	**Full Sample**			**Cross-Fitting**		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.2073	0.3806	0.0080	0.4638	0.5947	0.0088
T-Learner	0.0723	0.1776	0.0138	0.1165	0.2398	0.0194
IPW-Learner	0.3039	0.3774	0.0809	0.4673	0.2504	0.3661
DR-Learner	0.0602	0.1360	0.0223	0.1295	0.1570	0.0856
X-Learner	0.0628	0.1597	0.0147	0.0982	0.2164	0.0183
R-Learner	0.0573	0.1061	0.0326	0.1210	0.1656	0.0702

Table 4.8: Metrics for Low Noise on Outcome

Learner	**Full Sample**			**Cross-Fitting**		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.3070	0.4610	0.0278	0.6827	0.7291	0.0238
T-Learner	0.1208	0.1732	0.0652	0.2036	0.2447	0.1032
IPW-Learner	0.4284	0.3795	0.2021	0.9103	0.2612	0.8016
DR-Learner	0.1955	0.1422	0.1570	0.6020	0.1907	0.5408
X-Learner	0.1346	0.1601	0.0878	0.1890	0.2248	0.1046
R-Learner	0.2973	0.1254	0.2681	0.5587	0.1952	0.4933

Table 4.9: Metrics for High Noise on Outcome

The results regarding the impact of noise are depicted in Tables 4.8 and 4.9. As expected, all learners perform worse as the noise level increases, making it more challenging to estimate outcomes accurately from the data. In low-noise scenarios, the R-learner achieves the best performance in terms of both MSE and bias.

One might expect methods like the T-learner to perform better in these conditions, given the relative ease of estimating the response functions under treatment and control. However, the R-learner's lower bias gives it an edge in this context.

When noise levels are high, the R-learner's performance declines significantly, primarily due to increased variance. This may be attributed to greater variability in the numerator of the pseudo-outcomes, leading to more variance in the CATE estimates. Despite this, the R-learner still exhibits the lowest bias among the learners. Under high noise, the T-learner performs best with respect to MSE, mainly due to its lower variance compared to the other learners. As

noise complicates the estimation of response functions, the T-learner's simpler and more stable predictions of the CATE offer an advantage.

4.3 Impact of Propensity Score Variations

In this section, we explore how changes in the propensity score affect the performance of the learners.

Learner	**Full Sample**			**Cross-Fitting**		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.4009	0.5530	0.0136	0.7680	0.7908	0.0117
T-Learner	0.1335	0.2510	0.0251	0.2247	0.3438	0.0439
IPW-Learner	0.7351	0.5823	0.1060	0.7449	0.4001	0.4602
DR-Learner	0.1357	0.2097	0.0494	0.2573	0.2011	0.1803
X-Learner	0.0968	0.1934	0.0302	0.1580	0.2726	0.0386
R-Learner	0.2062	0.2350	0.0996	0.3676	0.3090	0.2076

Table 4.10: Metrics for Low Confounding

Learner	**Full Sample**			**Cross-Fitting**		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.9755	0.9114	0.0042	1.0821	0.9635	0.0026
T-Learner	0.5203	0.6278	0.0251	0.6849	0.7258	0.0448
IPW-Learner	2.0957	1.0819	0.0744	1.8309	0.9569	0.2243
DR-Learner	0.5241	0.6081	0.0427	0.6054	0.6070	0.1216
X-Learner	0.3010	0.4214	0.0319	0.4794	0.5529	0.0504
R-Learner	2.2556	0.8523	0.7432	1.9845	0.8067	0.9872

Table 4.11: Metrics for High Confounding

The results of varying levels of confounding are presented in Tables 4.10 and 4.11. We observe a decline in performance across all learners as the level of confounding increases, with the R-learner being particularly affected. In terms of bias, the DR-learner outperforms the T-learner, which is indirectly influenced by the propensity score through its impact on the distribution of treated and control groups. As confounding intensifies, the propensity score becomes less balanced, making it more challenging to accurately estimate the response functions. This imbalance might contributes to the increased bias observed in both the T-learner and S-learner. The X-learner outperforms all other learners, aligning with Künzel et al.'s [3] findings on imbalanced treatment assignments, where they compared the performances of the S-, T-, and X-learners.

The results regarding balanced and imbalanced treatment assignments are presented in Tables 4.12 and 4.13. These outcomes closely mirror those observed when varying the level of confounding, which is expected given the similar properties of the propensity score functions used in both scenarios. The main distinction lies in the fact that the propensity score used here

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.2239	0.3945	0.0126	0.5150	0.6270	0.0142
T-Learner	0.0796	0.1734	0.0235	0.1299	0.2369	0.0348
IPW-Learner	0.3152	0.3690	0.1038	0.5547	0.2467	0.4567
DR-Learner	0.0852	0.1355	0.0485	0.2252	0.1682	0.1762
X-Learner	0.0758	0.1574	0.0294	0.1153	0.2166	0.0357
R-Learner	0.1033	0.1084	0.0792	0.2087	0.1741	0.1549

Table 4.12: Metrics for Balanced Treatment Assignment

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.8198	0.8196	0.0094	1.0519	0.9437	0.0082
T-Learner	0.3430	0.4273	0.0495	0.6079	0.5375	0.1704
IPW-Learner	2.3656	1.2711	0.0777	3.6804	0.8018	2.6612
DR-Learner	0.4246	0.4652	0.0748	1.7013	0.3909	1.4669
X-Learner	0.1607	0.2507	0.0548	0.3166	0.3777	0.0960
R-Learner	5.8995	1.3886	1.6067	16.0551	2.0429	8.1967

Table 4.13: Metrics for Imbalanced Treatment Assignment

lacks the additional dependency on x_3, although this x_3 dependency was minimal, resulting in comparable propensity scores.

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.1736	0.3374	0.0124	0.4140	0.5504	0.0143
T-Learner	0.0732	0.1640	0.0232	0.1236	0.2239	0.0386
IPW-Learner	0.2094	0.2741	0.1079	0.5000	0.1727	0.4490
DR-Learner	0.0794	0.1255	0.0489	0.2170	0.1533	0.1766
X-Learner	0.0726	0.1520	0.0292	0.1081	0.2068	0.0349
R-Learner	0.0977	0.1012	0.0777	0.1916	0.1540	0.1504

Table 4.14: Metrics for Constant Treatment Assignment

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.8871	0.8647	0.0049	1.0923	0.9687	0.0025
T-Learner	0.5931	0.6077	0.0420	1.0105	0.8449	0.1026
IPW-Learner	2.5367	1.3193	0.0719	2.0520	0.7581	1.1736
DR-Learner	0.6501	0.5859	0.0651	1.1295	0.5045	0.7160
X-Learner	0.1564	0.2659	0.0478	0.3844	0.4862	0.0695
R-Learner	7.2336	1.2459	0.8820	9.2045	1.4306	2.7940

Table 4.15: Metrics for Complex Treatment Assignment

The results for constant treatment assignment and complex treatment assignment are presented in Tables 4.14 and 4.15. Under constant treatment, we observe a slight improvement compared

to the balanced case, which is expected due to the slight imbalance introduced by the balanced case's propensity score dependency on x_1. However, since this dependency is minimal, it closely resembles the constant treatment scenario.

In the complex treatment assignment case, the results are similar to the imbalanced scenario, with the key difference being a stronger dependency on x_3, which increases the complexity of the propensity score. Notably, there is a significant increase in bias for the R-learner.

4.4 Impact of CATE Variations

In this section, we examine how changes in the CATE affect the performance of the learners.

Learner	**Full Sample**			**Cross-Fitting**		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.1065	0.3145	0.0072	0.3099	0.5459	0.0118
T-Learner	0.0154	0.0429	0.0133	0.0276	0.0577	0.0242
IPW-Learner	0.2083	0.3033	0.0974	0.3928	0.0866	0.3799
DR-Learner	0.0378	0.0372	0.0360	0.1528	0.0391	0.1505
X-Learner	0.0203	0.0263	0.0193	0.0277	0.0394	0.0258
R-Learner	0.0670	0.0343	0.0653	0.1356	0.0662	0.1299

Table 4.16: Metrics for Independent CATE

Learner	**Full Sample**			**Cross-Fitting**		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.3701	0.4845	0.0143	0.6731	0.6822	0.0126
T-Learner	0.2064	0.3337	0.0288	0.2721	0.3873	0.0350
IPW-Learner	0.4081	0.4320	0.1180	0.6300	0.3331	0.4562
DR-Learner	0.1866	0.2822	0.0588	0.3234	0.2876	0.1923
X-Learner	0.1869	0.3082	0.0351	0.2399	0.3574	0.0377
R-Learner	0.1892	0.2427	0.0942	0.3087	0.2969	0.1679

Table 4.17: Metrics for Highly Dependent CATE

The results of varying the number of covariates on which the CATE depends are presented in Tables 4.16 and 4.17. When the treatment effect is constant and non-zero, the T-learner achieves the best MSE performance. This might be attributed to the clear distinction between the treatment and control groups, which is independent of covariates, allowing the T-learner to effectively estimate the response functions for both groups. Although the X-learner performs best in terms of bias, it shows slightly higher variance compared to the T-learner. When the CATE depends on 15 covariates, the X-, R-, and DR-learners outperform the others in terms of MSE. The R-learner achieves the lowest bias, while the X-learner exhibits the lowest variance.

The results of varying levels of complexity in the CATE are depicted in Tables 4.18 and 4.19. In the case of no treatment effect, the S-learner performs best. This could be explained by the S-

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.0002	0.0065	0.0001	0.0002	0.0048	0.0002
T-Learner	0.0154	0.0429	0.0133	0.0276	0.0577	0.0242
IPW-Learner	0.0849	0.0627	0.0763	0.2852	0.0707	0.2771
DR-Learner	0.0379	0.0372	0.0360	0.1528	0.0391	0.1504
X-Learner	0.0203	0.0263	0.0193	0.0278	0.0394	0.0258
R-Learner	0.0691	0.0455	0.0661	0.1347	0.0672	0.1289

Table 4.18: Metrics for No Treatment Effect

Learner	Full Sample			Cross-Fitting		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	7.9481	1.7521	0.1144	12.1447	2.1828	0.0616
T-Learner	5.8099	1.5650	0.2358	9.1505	2.0397	0.1946
IPW-Learner	5.3350	1.4413	0.3501	5.9806	1.5026	1.2221
DR-Learner	4.2130	1.2923	0.2839	5.2991	1.4244	0.5952
X-Learner	5.2084	1.4734	0.2289	7.8962	1.8736	0.1925
R-Learner	3.0183	1.0833	0.3814	5.3017	1.4294	0.5616

Table 4.19: Metrics for Complex Treatment Effect

learner's approach of modeling the outcome as a single function of both covariates and treatment status, fitting one model for the entire dataset. In contrast, the T-learner fits separate models for the treated and control groups, introducing unnecessary complexity when the treatment effect is zero. The S-learner naturally captures the absence of a treatment effect within a single model, avoiding the additional complexity of the T-learner, which attempts to differentiate between two groups even when their outcomes are essentially the same. The IPW-learner and R-learner perform the worst in detecting no treatment effect.

When examining a complex treatment effect, the R-learner performs best in terms of both MSE and bias. This might be due to the increased difficulty in estimating the response functions caused by the complexity of the treatment effect, leading learners that rely more on the propensity score to perform better. Interestingly, for the S-, T-, and X-learners, cross-fitting exhibits less variance. This reduction in variance might be attributed to the tendency of complex response functions to overfit, where splitting the data into smaller parts through cross-fitting helps mitigate this issue. However, despite the reduction in variance, bias increases, leading to an overall decline in performance for the cross-fitting variants.

4.5 Impact of Random Forest Parameters and Cross-Fitting Configurations

In this section, we explore how varying the parameters of the random forest and experimenting with different cross-fitting configurations can influence the performance of various learners.

Learner	**Full Sample**			**Cross-Fitting**		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.2906	0.4517	0.0092	0.5955	0.6732	0.0106
T-Learner	0.1018	0.2161	0.0188	0.1588	0.2835	0.0266
IPW-Learner	0.2307	0.2942	0.0866	0.4230	0.2740	0.3023
DR-Learner	0.0916	0.1557	0.0449	0.2006	0.1964	0.1341
X-Learner	0.0912	0.1939	0.0245	0.1373	0.2583	0.0261
R-Learner	0.0952	0.1483	0.0542	0.1900	0.2071	0.1161

Table 4.20: Metrics for Shallow Trees

Learner	**Full Sample**			**Cross-Fitting**		
	MSE	**Bias**	**Variance**	**MSE**	**Bias**	**Variance**
S-Learner	0.1224	0.1871	0.0615	0.3371	0.4383	0.0864
T-Learner	0.0855	0.1034	0.0654	0.2015	0.1753	0.1514
IPW-Learner	0.4789	0.4682	0.1588	1.4058	0.2405	1.3104
DR-Learner	0.0967	0.0841	0.0828	0.3684	0.1114	0.3472
X-Learner	0.0706	0.0927	0.0545	0.1075	0.1337	0.0765
R-Learner	0.1608	0.0817	0.1482	0.3308	0.1042	0.3115

Table 4.21: Metrics for Deep Trees

The results of varying random forest parameters are presented in Tables 4.20 and 4.21. We observe that the S-, T-, and X-learners perform better with deeper trees, while the DR-, R-, and X-learners perform better with shallower trees. This difference may be attributed to the reduced regularization in deeper trees, which increases the risk of overfitting. This suggests potential underfitting in the response function estimation and overfitting in the propensity score function. Notably, the R-learner shows a significant increase in variance with deep trees, indicating sensitivity to overfitting, while the IPW-learner also exhibits increased variance, though to a lesser extent.

The results for different cross-fitting settings are presented in Table 4.22. When comparing three-fold to five-fold cross-fitting, all learners perform better with three-fold cross-fitting, though still not as well as the full sample estimators. This finding aligns with our earlier observation that using less data generally leads to less accurate and less stable estimates of the nuisance functions. Additionally, we observe that mean cross-fitting increases variance across the board compared to median cross-fitting, though it reduces bias. The higher variance could be attributed to the mean's sensitivity to outliers, while the bias reduction suggests that the mean

Learner	3-Fold Cross-Fitting			Mean Cross-Fitting		
	MSE	Bias	Variance	MSE	Bias	Variance
S-Learner	0.4389	0.5683	0.0119	0.5237	0.6332	0.0141
T-Learner	0.1313	0.2442	0.0301	0.1324	0.2392	0.0355
IPW-Learner	0.4544	0.2362	0.3634	0.5548	0.2510	0.4535
DR-Learner	0.1770	0.1559	0.1338	0.2241	0.1669	0.1754
X-Learner	0.1103	0.2181	0.0292	0.1157	0.2171	0.0357
R-Learner	0.1748	0.1676	0.1241	0.2085	0.1747	0.1537

Table 4.22: Metrics for Different Cross-Fitting Settings

better captures variation in the data, offering a more accurate reflection of the true average effect. However, this comes at the cost of increased instability in the estimates.

4.6 Discussion

The primary objective of this study was to evaluate various meta-learners and determine whether cross-fitting improves the accuracy of CATE estimates. Contrary to expectations, our findings indicate that cross-fitting consistently reduces the accuracy of estimates across different scenarios. However, as the sample size increased, the accuracy gap between full-sample and cross-fitting estimates narrowed. This suggests that with a larger sample size, cross-fitting might yield different results, warranting further investigation. Okasa's [7] simulation study supports this possibility, showing that increased sample sizes can improve cross-fitting performance in certain cases. Given the differences in data generation methods between our study and Okasa's, it remains an open question whether similar results would hold in our settings.

These results were somewhat surprising, particularly in light of Jacob's [6] study, which found that cross-fitting improved estimation accuracy, even with a sample size of 2000, under specific conditions. The discrepancy between our findings and Jacob's may be attributed to differences in data generation, suggesting that the impact of cross-fitting is likely dependent on the data's properties.

Regarding the performance of the learners, we observed that when the number of covariates was small, the S-, T-, and X-learners performed best, whereas with a larger number of covariates, the DR- and R-learners excelled. A consistent trend across scenarios was the poor performance of the IPW-learner, leading us to advise against its use based on our simulation results. Another key observation is that the S-learner performs well in the absence of a treatment effect, which might seem advantageous. However, across different scenarios, the S-learner exhibited high bias and low variance, indicating its limited usefulness in detecting treatment effects, as its CATE estimates tend to be biased toward zero.

We also found that as covariate correlation increases, the performance of all learners deteri-

orates. In high-noise settings, the X- and T-learners outperformed others, while in low-noise settings, the R-, DR-, and X-learners were most effective. The X-learner was particularly strong in scenarios with imbalanced treatment assignment, while in cases of complex treatment effects, the R-learner performed best in terms of bias, and the X-learner excelled in terms of variance.

Varying the random forest parameters revealed that deeper trees (with less regularization) favored the X-, DR-, and T-learners, while shallower trees benefitted the R-, DR-, and X-learners. When we reduced the number of folds in cross-fitting, accuracy improved compared to five-fold cross-fitting, though it remained less accurate than full-sample estimations.

In conclusion, while this study provided valuable insights into the performance of meta-learners with and without cross-fitting, some results were difficult to explain definitively. Future research could benefit from examining not only the MSE, bias, and variance of CATE estimates but also those of the nuisance functions themselves. This approach could shed more light on the underlying reasons for the performance differences observed among learners. Our simulation yielded intriguing results, some of which align with existing studies, while others do not, suggesting that the performance of meta-learners—whether or not cross-fitting is applied—depends on the specific characteristics of the data being analyzed.

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Appendix A

Nederlandse Samenvatting

Het belangrijkste doel van deze scriptie was om de prestaties van verschillende meta-learners te evalueren—specifiek de S-learner, T-learner, Inverse Probability Weighting (IPW) learner, Doubly Robust (DR) learner, X-learner en R-learner—bij het schatten van het Conditional Average Treatment Effect (CATE). Daarnaast had het onderzoek als doel om te bepalen of het toepassen van cross-fitting op deze meta-learners de nauwkeurigheid en robuustheid van de CATE-schattingen in verschillende scenario's zou kunnen verbeteren.

De studie begon met het vaststellen van de fundamentele concepten van causal inference, met een bijzondere nadruk op het potential outcomes-framework. Dit framework vormt de theoretische basis voor het definiëren van causale effecten, met name de CATE. Om de CATE nauwkeurig te schatten, moeten er aan verschillende belangrijke assumpties worden voldaan, waaronder de Stable Unit Treatment Value Assumption (SUTVA), positiviteit, en conditionele uitwisselbaarheid. Deze aannames zijn cruciaal omdat ze de identificatie van de CATE uit de waargenomen data mogelijk maken, waardoor de causale effecten correct kunnen worden afgeleid.

Op basis van deze basis beschreef het methodologiehoofdstuk de meta-learners die in de studie zijn gebruikt, elk met zijn sterke en zwakke punten, afhankelijk van de toepassingscontext. Ook het cross-fitting-algoritme werd in detail uitgelegd, waarbij de rol van het opsplitsen van data in delen om overfitting te verminderen en de betrouwbaarheid van schattingen te verbeteren, werd benadrukt.

Vervolgens beschreef de scriptie het simulatiedesign dat werd gebruikt om data te genereren en de prestaties van de meta-learners te evalueren. Deze simulaties waren zorgvuldig geconstrueerd om real-world scenario's na te bootsen, met variërende steekproefgroottes, covariatenruimtes, correlatiestructuren, ruisniveaus en behandelingsverdelingen. Deze diverse scenario's maakten een uitgebreide beoordeling mogelijk van hoe goed elke learner presteert onder ver-

schillende omstandigheden. De prestaties van de meta-learners werden geëvalueerd aan de hand van drie belangrijke meetwaarden: gemiddelde Mean Squared Error, gemiddelde absolute bias en gemiddelde variantie.

De resultaten van de simulaties leverden verschillende belangrijke inzichten op. Verrassend genoeg verbeterde cross-fitting de nauwkeurigheid van de CATE-schattingen niet; in feite leidde het consequent tot slechtere prestaties in vergelijking met schattingen op basis van de volledige steekproef. De studie wees ook uit dat de prestaties van de meta-learners aanzienlijk varieerden, afhankelijk van het scenario. De S-, T- en X-learners presteerden het beste bij een klein aantal covariaten, terwijl de DR- en R-learners uitblonken in omgevingen met hoge dimensies. De IPW-learner presteerde consistent slecht in alle scenario's, wat leidde tot de aanbeveling om het gebruik van deze learner op basis van onze simulatieresultaten te vermijden. De S-learner, hoewel effectief in scenario's zonder behandelingseffect, vertoonde hoge bias en lage variantie in verschillende scenario's, wat wijst op de beperkte bruikbaarheid ervan bij het detecteren van behandelingseffecten, aangezien de CATE-schattingen de neiging hebben om naar nul te worden vertekend.

De resultaten gaven ook aan dat naarmate de covariaten correlatie toenam, de prestaties van alle learners verslechterden. In scenario's met hoge ruis presteerden de X- en T-learners beter dan de andere, terwijl in scenario's met lage ruis de R-, DR- en X-learners het meest effectief waren. De X-learner presteerde bijzonder goed in scenario's met een onevenwichtige toewijzing van behandelingen, terwijl de R-learner uitblonk in gevallen met complexe behandelingseffecten, waarbij hij het beste presteerde in termen van bias.

Conclusie bood deze scriptie waardevolle inzichten in de prestaties van meta-learners met en zonder cross-fitting. Hoewel cross-fitting de schattingen niet verbeterde zoals verwacht, benadrukte de studie het belang van het begrijpen van de specifieke eigenschappen van de data bij het kiezen van de juiste methode voor CATE-schatting.